

Basic Principles of Medicine II

Module: Nervous System and Behavioral Science | Lecture NB 23

Development of the Skull, Face and Paranasal Sinuses

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Recommended reading

- Gary C. Schoenwolf, PhD; Steven B. Bleyl, MD, PhD; Philip R. Brauer, PhD; Philippa H. Francis-West, PhD
[Larsen's Human Embryology](#), 17, 419-462

Images from:

- Gary C. Schoenwolf, PhD; Steven B. Bleyl, MD, PhD; Philip R. Brauer, PhD; Philippa H. Francis-West, PhD
[Larsen's Human Embryology](#), 17, 419-462
- Richard L. Drake, PhD, FAAA; A. Wayne Vogl, PhD, FAAA; Adam W.M. Mitchell, MB BS, FRCS, FRCR
[Gray's Anatomy for Students](#), 8, 817-1114.e28
- T.V.N. (Vid) Persaud, MD, PhD, DSc, FRCPath (Lond.), FAAA; Mark G. Torchia, MSc, PhD
[The Developing Human](#), 14, 317-334

Recommended Reading

Moore KL, Persaud TVN, Torchia MG. The developing human: Clinically oriented embryology. 11th ed. Elsevier. Chapter 9, Chapter 14, Chapter 17

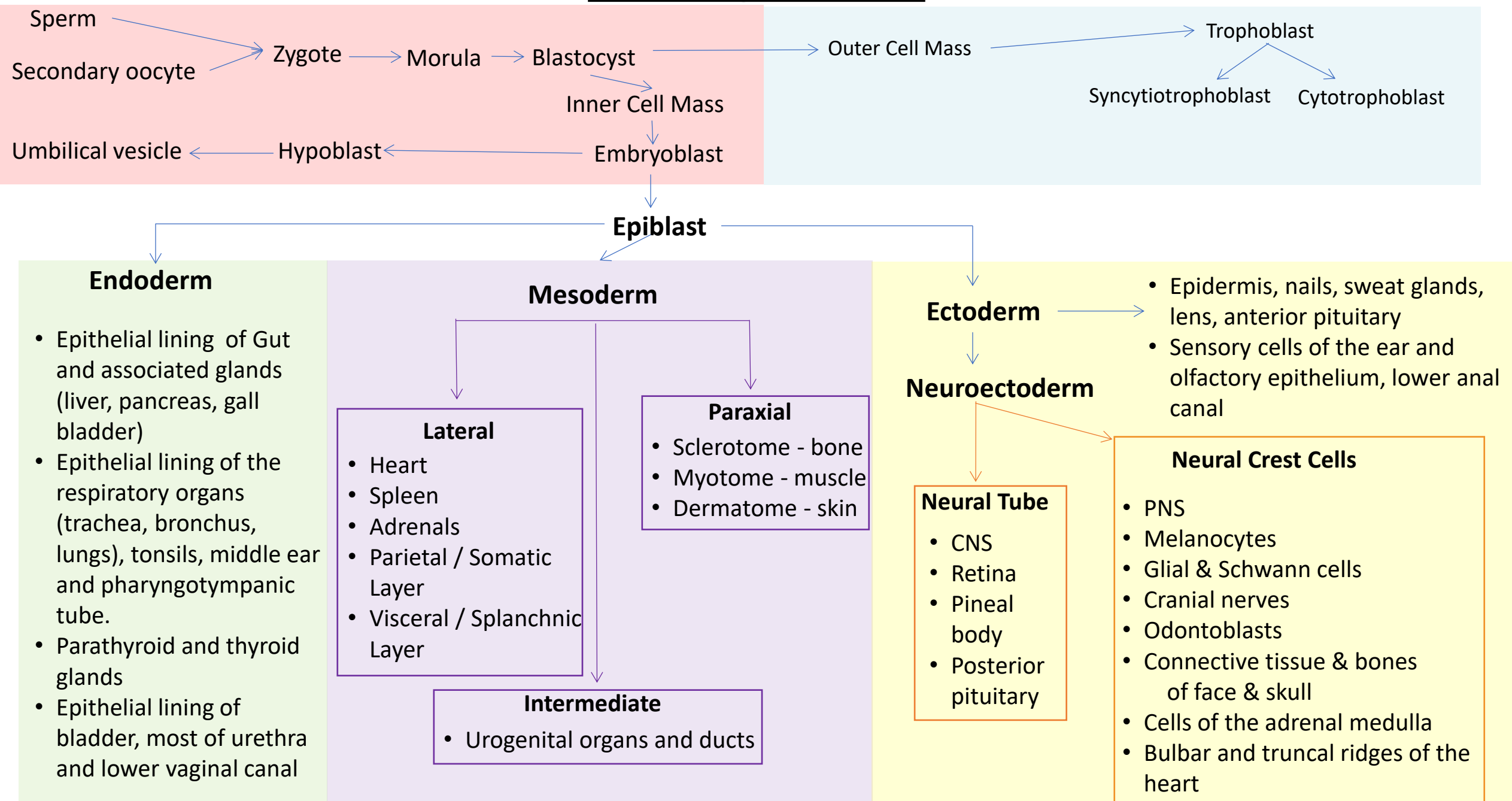


Pay special attention to the Blue boxes for the clinical correlations for this topic.

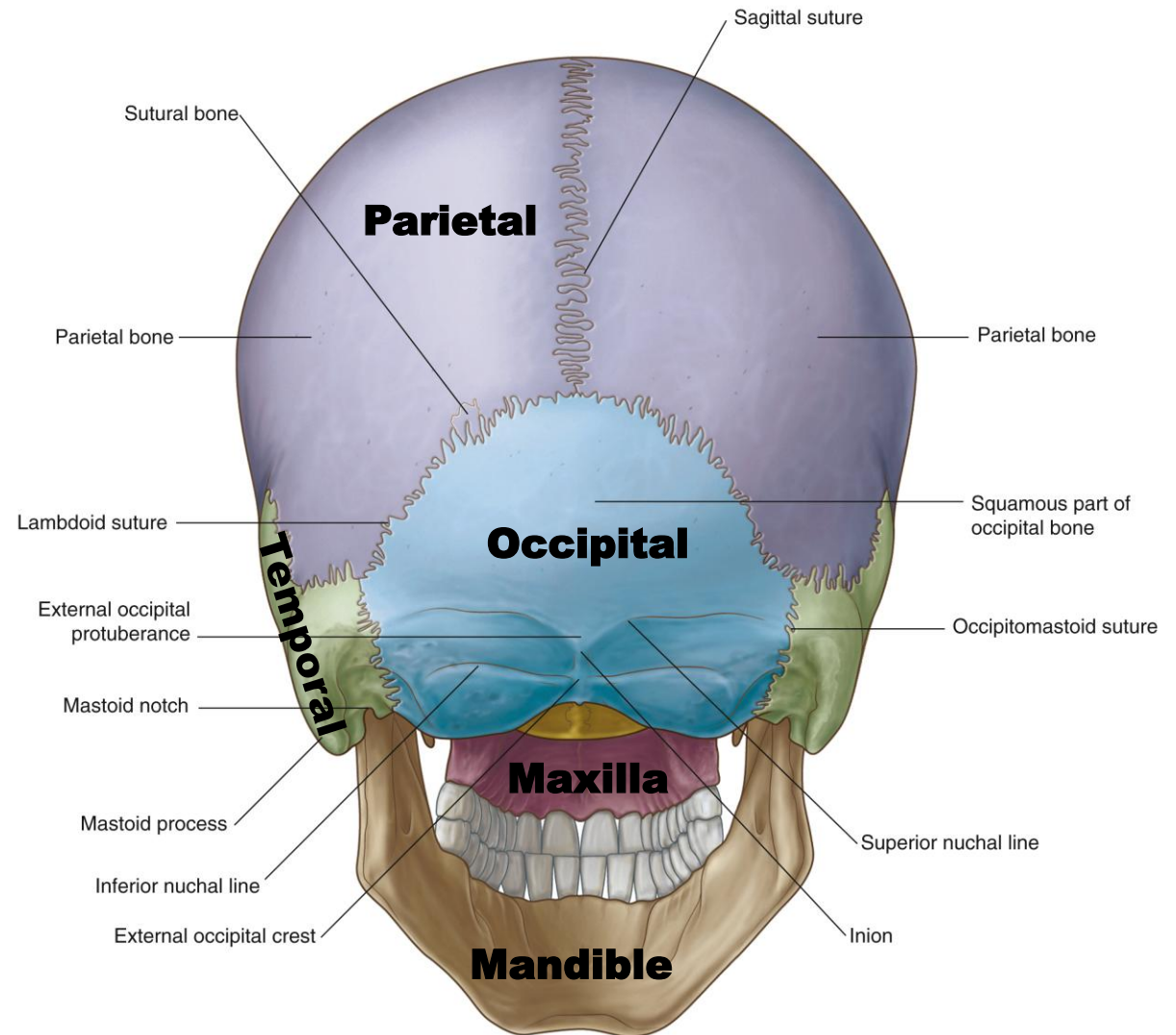
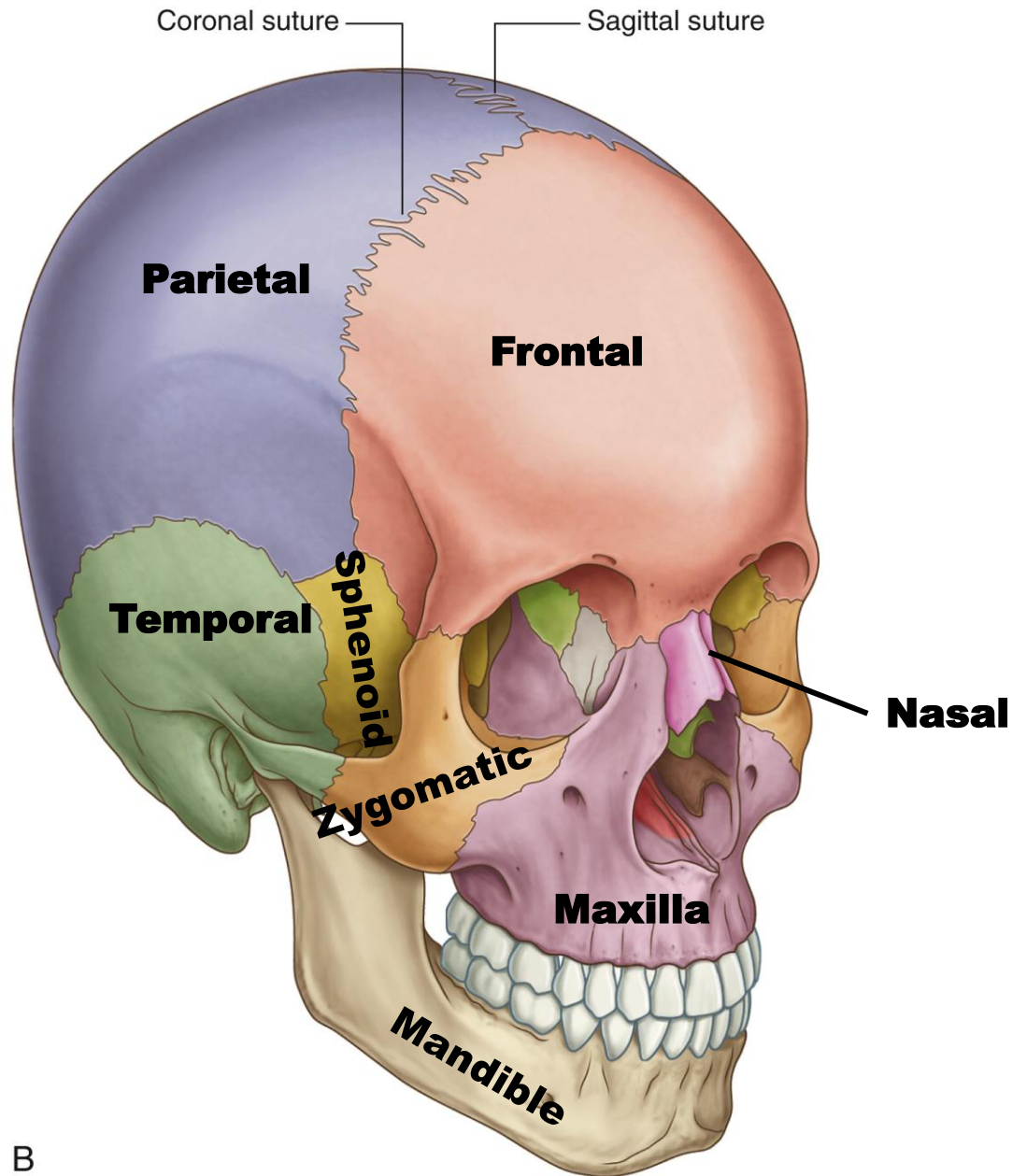
Objectives

SOM.MK.I.BPM2.1.NB.1.ANAT.0204	Describe the development of the viscerocranium and neurocranium
SOM.MK.I.BPM2.1.NB.1.ANAT.0205	Describe the closure of the sutures of the skull
SOM.MK.I.BPM2.1.NB.1.ANAT.0206	Describe the development of scaphocephaly, brachycephaly, plagiocephaly, trigonocephaly and cranium bifidum
SOM.MK.I.BPM2.1.NB.1.ANAT.0211	Describe the development of the face and palate from the five primordial prominences
SOM.MK.I.BPM2.1.NB.1.ANAT.0212	Describe the different forms of cleft lip and palate and the developmental processes that fail leading to these defects
SOM.MK.I.BPM2.1.NB.1.ANAT.0213	Describe the development of the nasal cavity
SOM.MK.I.BPM2.1.NB.1.ANAT.0214	Describe the development of the paranasal sinuses.

The Big picture



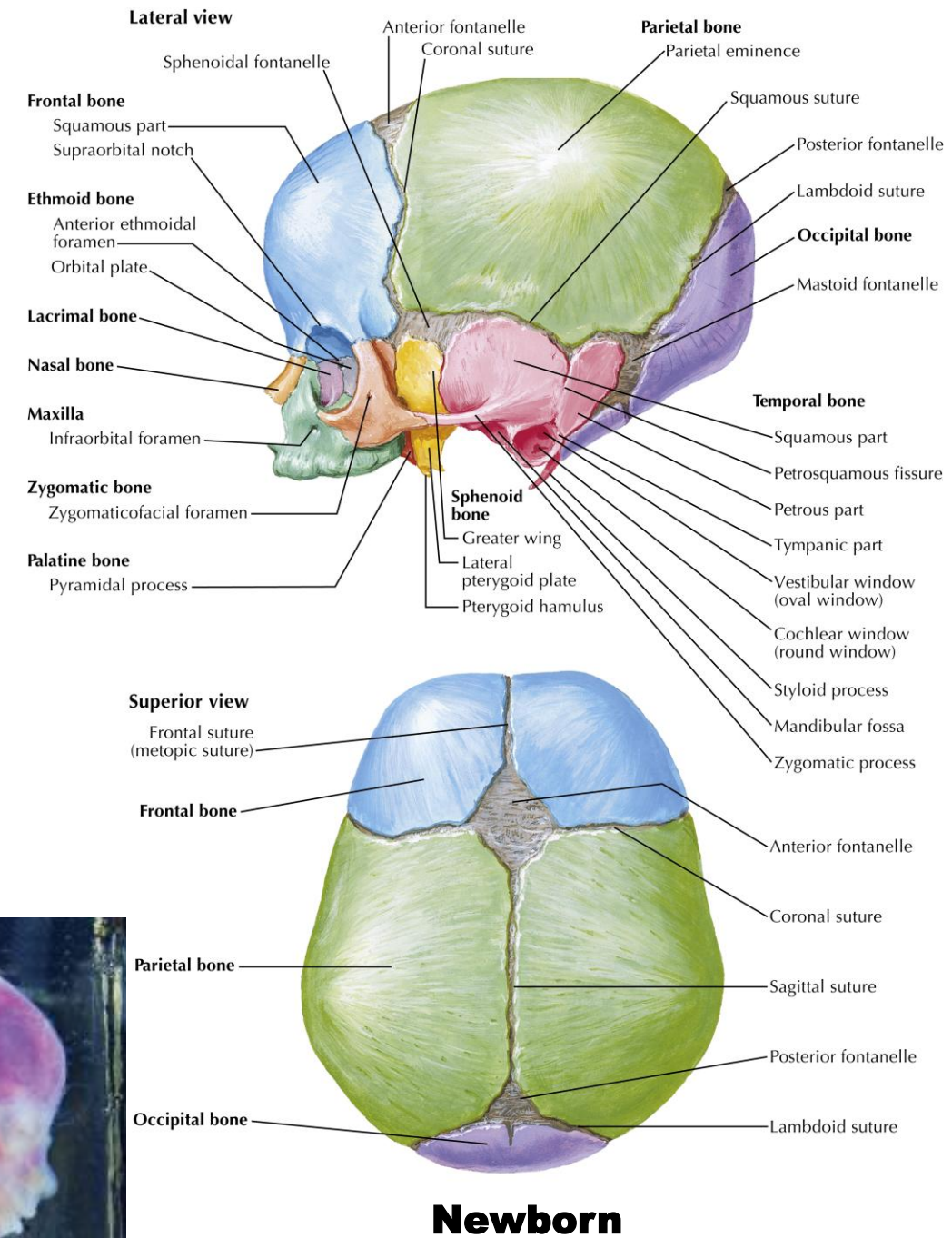
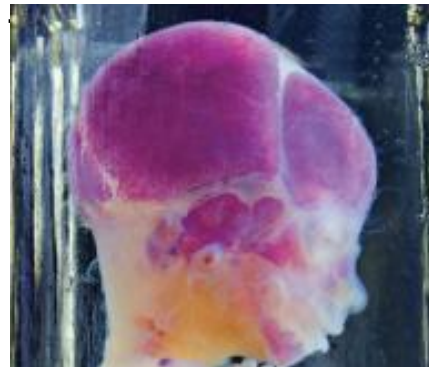
Adult structures



Development of the Skull

- The skull develops from the **mesenchyme** surrounding the developing brain
 - Neural crest derived**
- Skull has two parts
 - Neurocranium – bones that protect the brain and sensory organs (cranial base and cranial vault)
 - Viscerocranium – bones of the face and pharyngeal arches
- Each part of the bony skull is formed by mechanisms – **intramembranous and endochondral ossification**

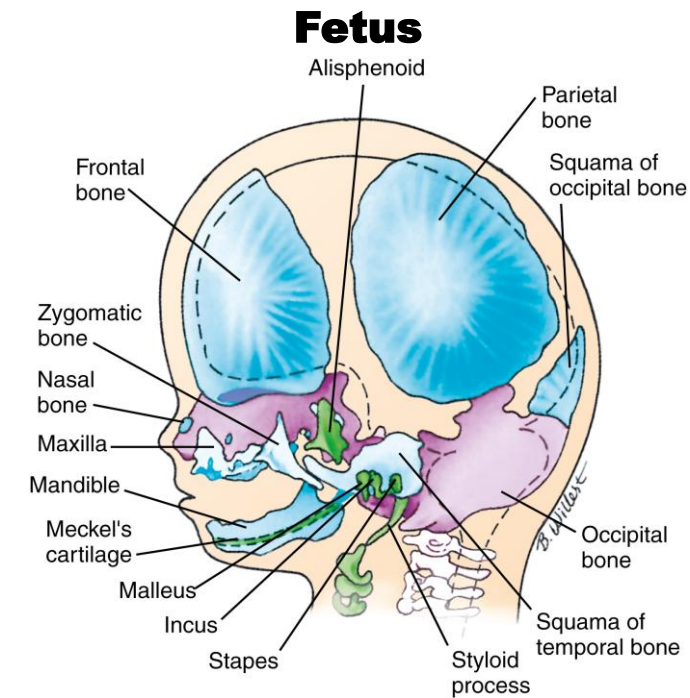
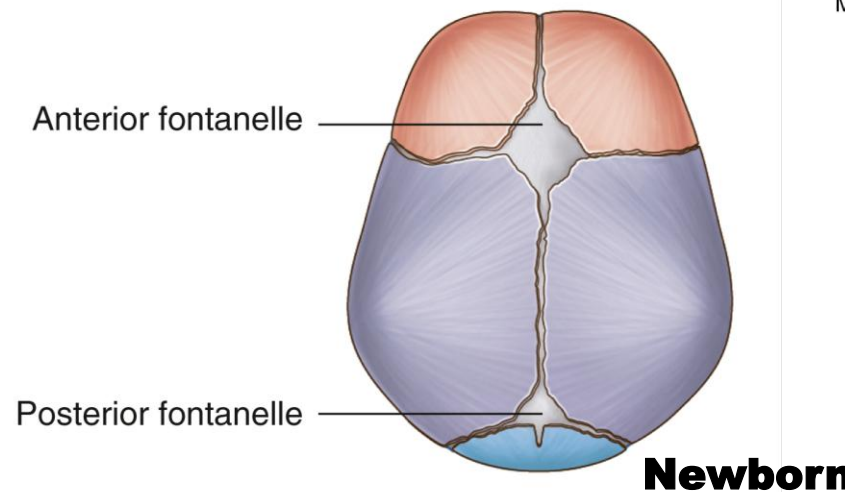
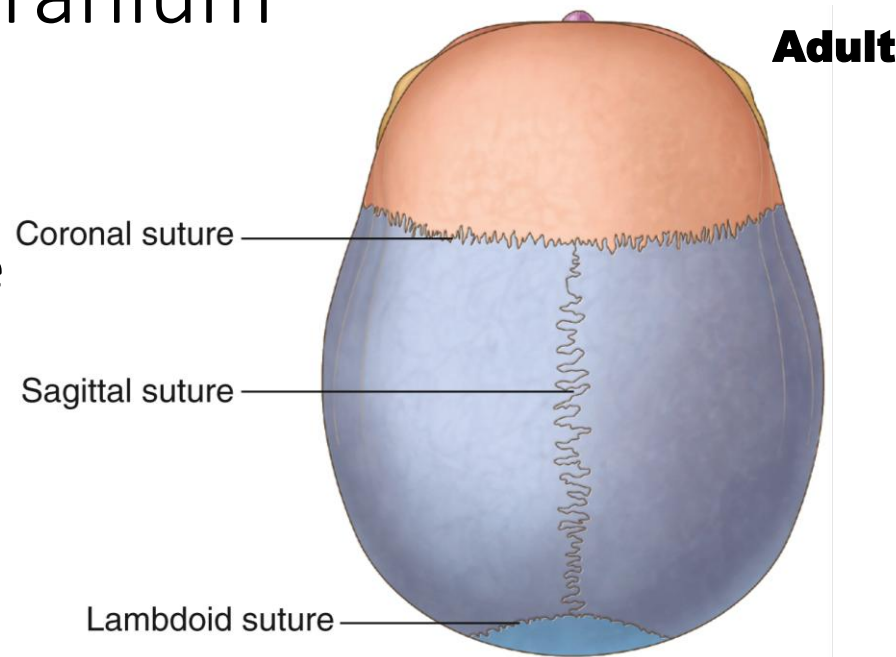
Fetus



Membranous Neurocranium

Calvaria (cranial vault)

- **Ossification centers** within the **mesenchyme** around the developing brain
- At the sides and top of the brain, forming the cranial vault
- **Parietal and frontal bones**
- The flat bones of the **cranial vault** are separated by **joints = sutures** and **membrane filled spaces = fontanelles**



Sutures - Joints of the skull

Frontal (Metopic suture):

- Extends from the top of the head **down the middle of the forehead**, toward the nose
- **2 frontal bone plates meet**

Coronal suture:

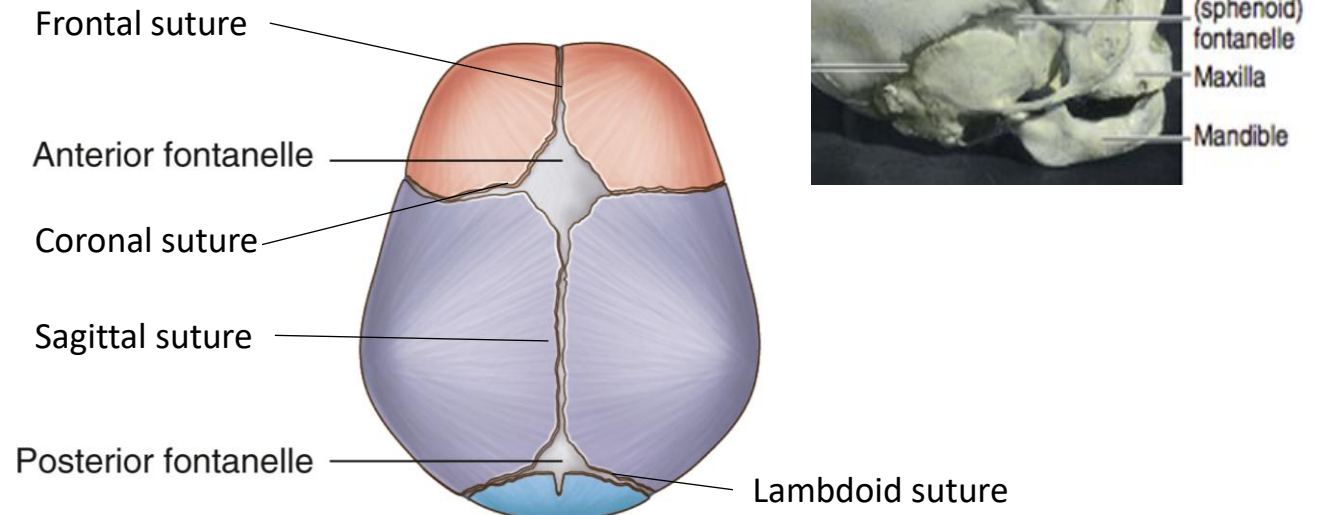
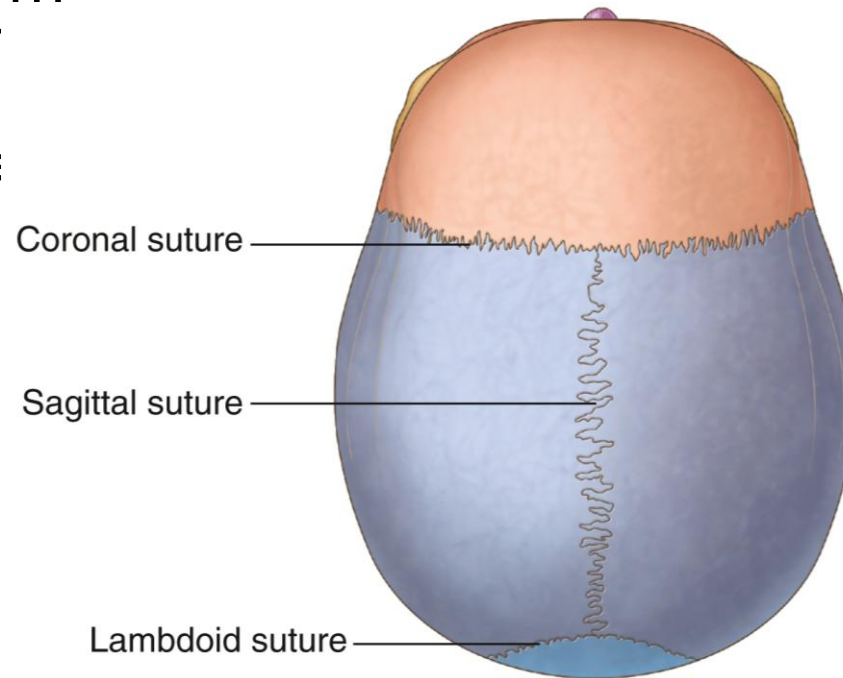
- Extends from ear to ear
- Each **frontal bone plate meets** with a **parietal bone plate**

Sagittal suture:

- Extends from the top of the head down to the back
- The **2 parietal bone plates meet**

Lambdoid suture:

- This extends across the back of the head
- Each **parietal bone plate meets** the **occipital bone plate**



Fontanelles – spaces that allow for growth

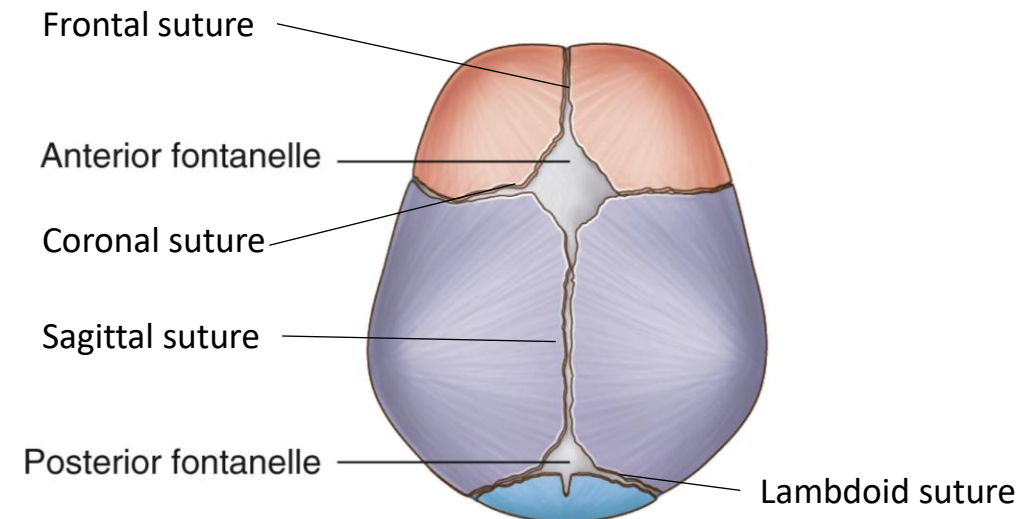
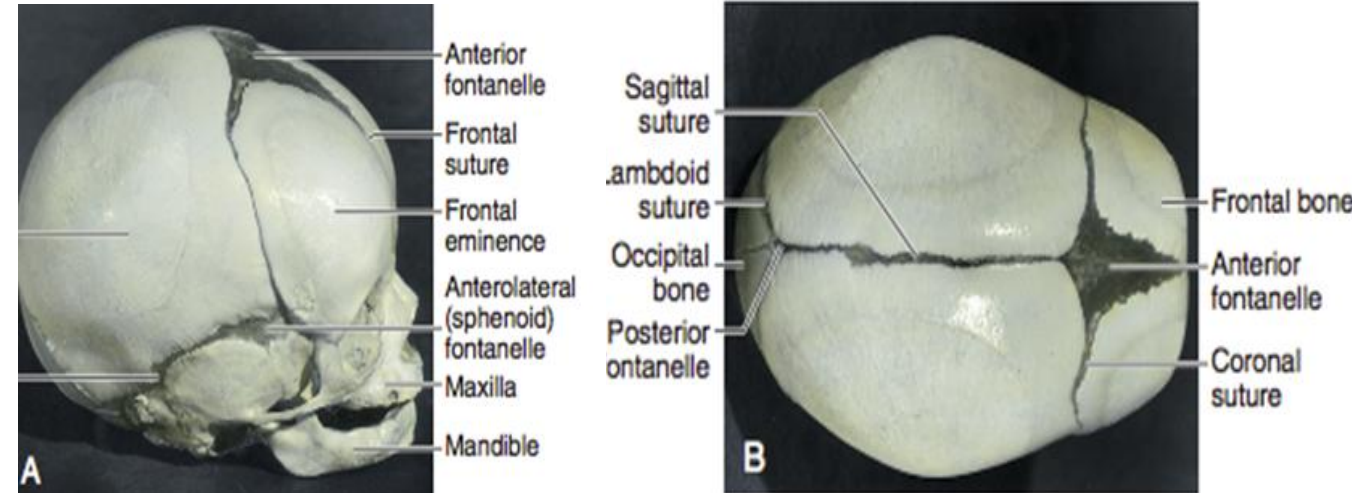
Two main fontanelles

Anterior:

- the junction where the two **frontal and parietal bones meet**
- closes at 18-24 months
- can be used to assess intracranial pressure due to e.g., intracranial masses
- **a sunken fontanelle is generally a sign of dehydration**

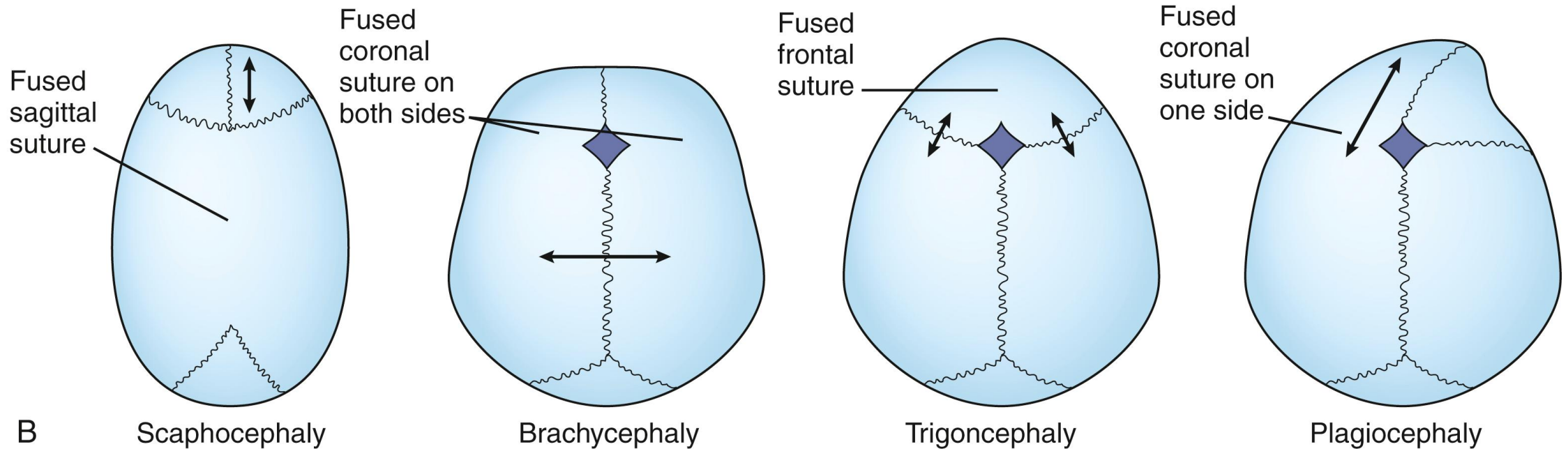
Posterior:

- the junction where the **2 parietal bones meet the occipital bone**
- generally closes before the anterior fontanelle at about 3 months



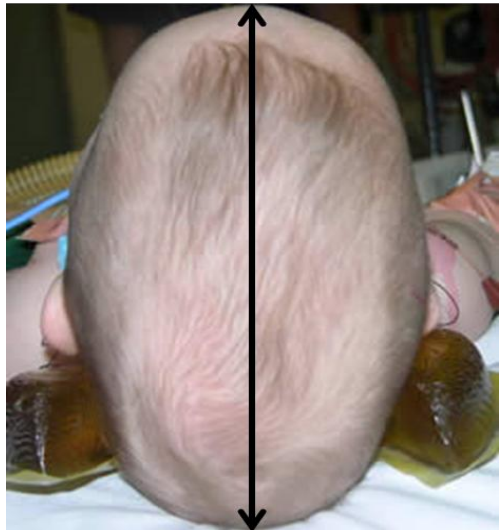
Clinical correlates - Craniosynostosis

One or more of the fibrous sutures in an infant skull **fuses prematurely** thereby changing the growth pattern of the skull



Scaphocephaly

- Caused by the premature closure of the **sagittal suture**
- **Most frequent** type of craniosynostosis
- Anterior posterior elongation
- On physical examination, there is a long narrow shaped head with prominent frontal and occipital regions
- CT scan shows long narrow shape of the head with bossing of the frontal and occipital regions



Scaphocephaly - Premature closure of the sagittal sutures.

Brachycephaly

- Craniosynostosis due to the premature closure of both **coronal sutures**
- On physical exam, there might be flattening of the frontal and occipital regions



Brachycephaly - Premature closure of the coronal sutures

Trigonocephaly

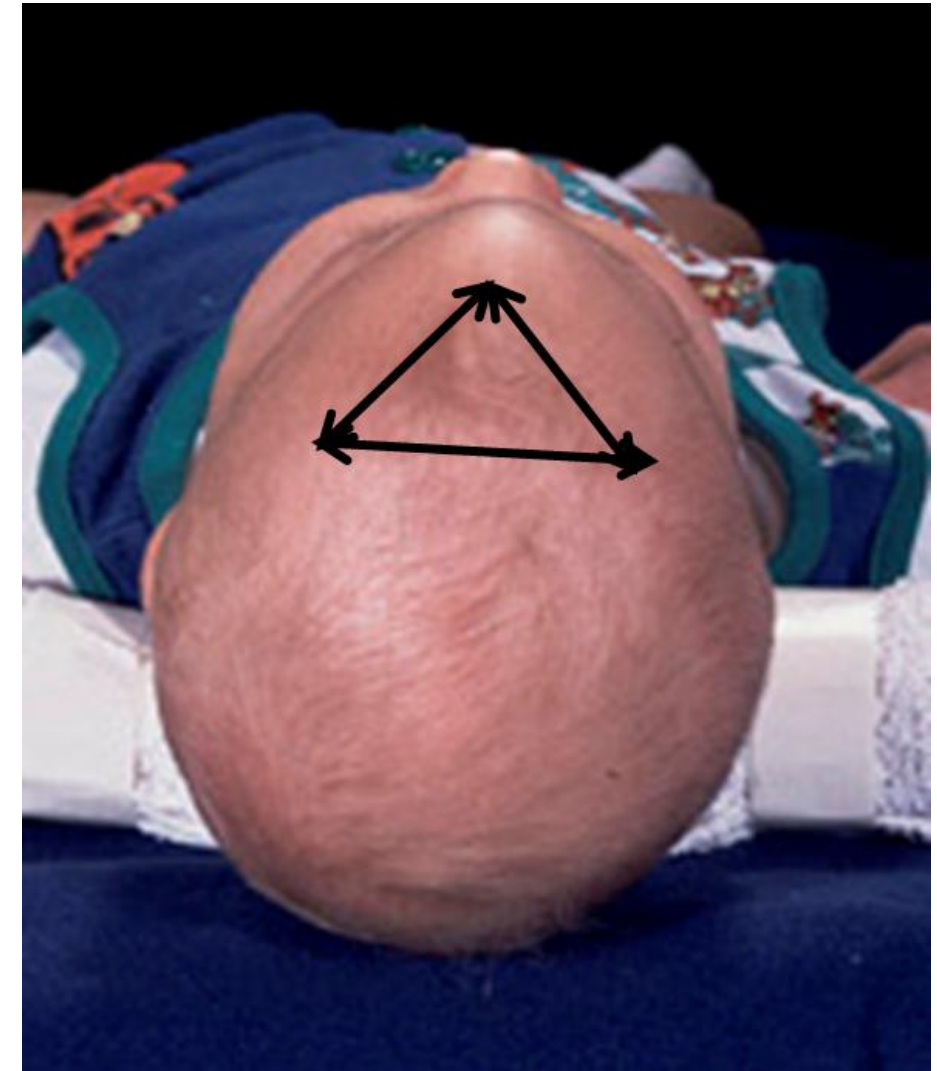
Premature closure of the frontal (metopic) suture

Physical examination:

- Noticeable ridge extending along center of forehead
- Forehead appears narrow
- Eyes may be close together
- Front of the skull may be pointed and triangular

Clinical symptoms:

- In extreme cases;
 - Developmental delays
 - Learning and behavioral problems
 - Vision defects



Trigonocephaly - Premature closure of the frontal suture

Plagiocephaly

- **Asymmetrical** premature closure of sutures

Clinical features

- Missing, full, or bulging "soft spot" (anterior fontanel) on the newborn's skull
- Bony ridges along the affected sutures
- Facial abnormalities including bulging forehead and brow on one side, uneven cheekbones, eye sockets, or lower jaw
- Prominent blood vessels in the scalp
- Poor feeding or projectile vomiting
- Seizures

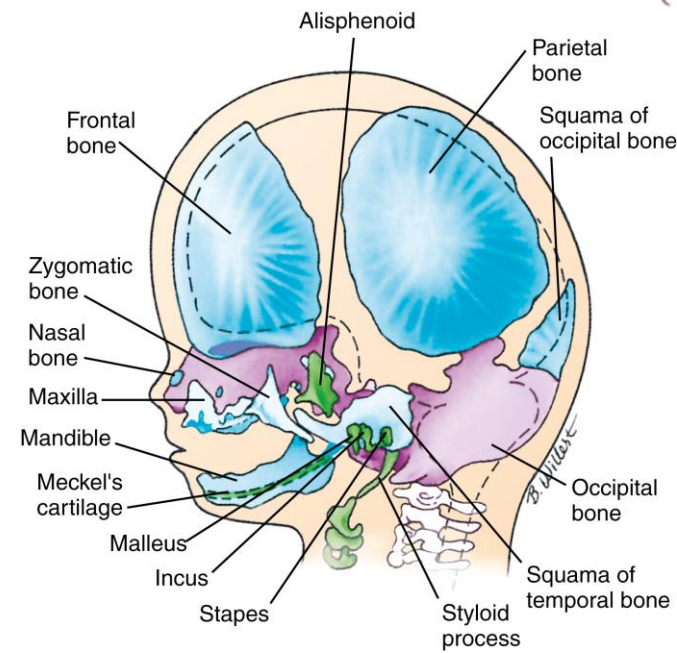
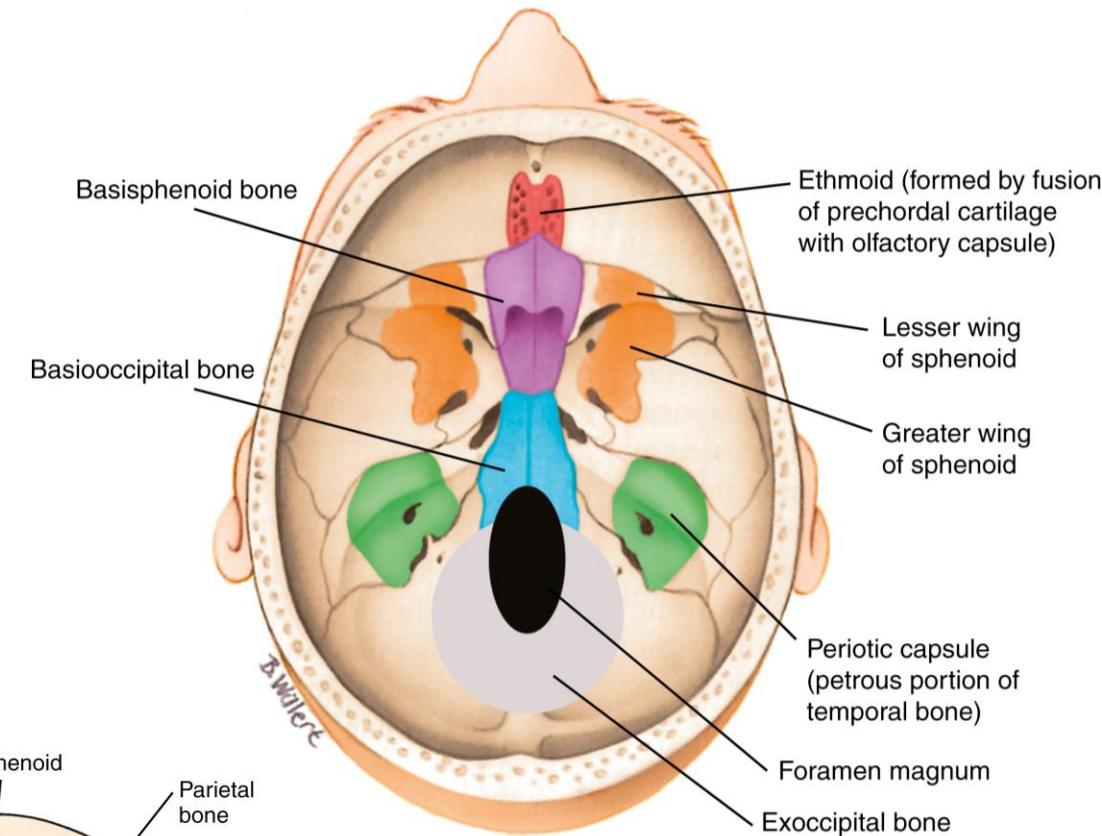


Plagiocephaly - Premature closure of the coronal suture on one side of the skull

Cartilaginous Neurocranium

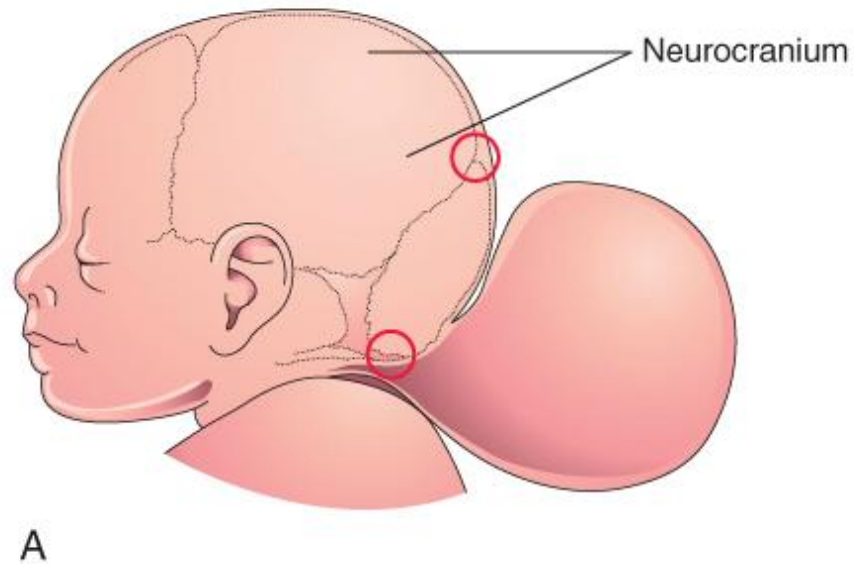
Cranial base

- bones in the base of the cranium, **the occipital bone, sphenoid, and ethmoid bone, petrous part of temporal bone**
- Endochondral ossification
- Cartilage derived from various somites



Cranium Bifidum

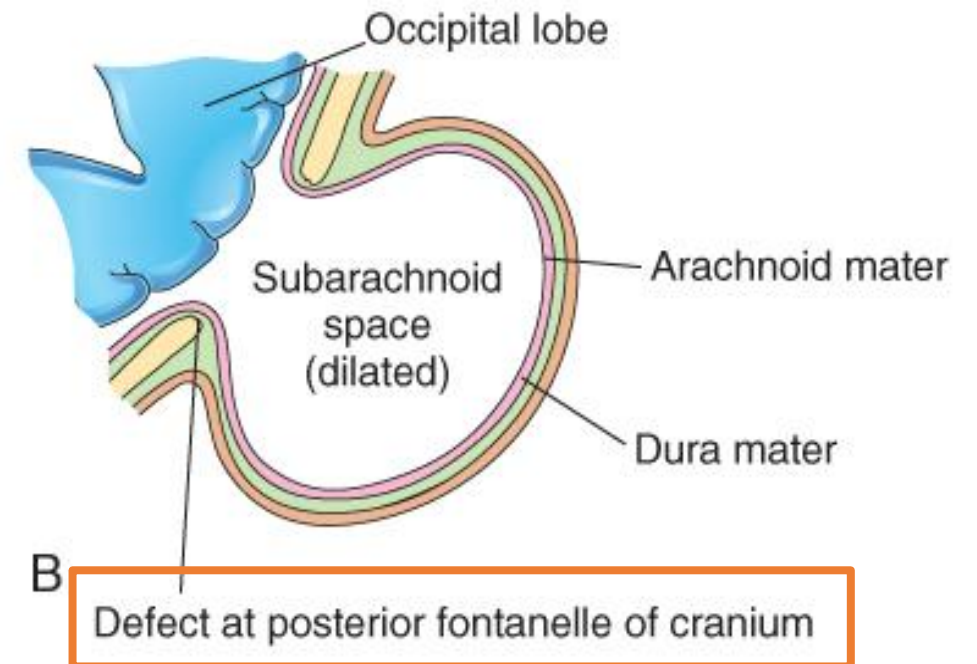
- Defect in the formation of cranial vault usually in the midline
- **Often in the occipital bone including foramen magnum** but can occur in the frontal and nasal bones
- Leads to encephalocele - herniation of intracranial contents
- **Various types** depending on what herniates
- Approximately 1 in 2000 births



Cranium Bifidum

Cranium Bifidum with Meningocele

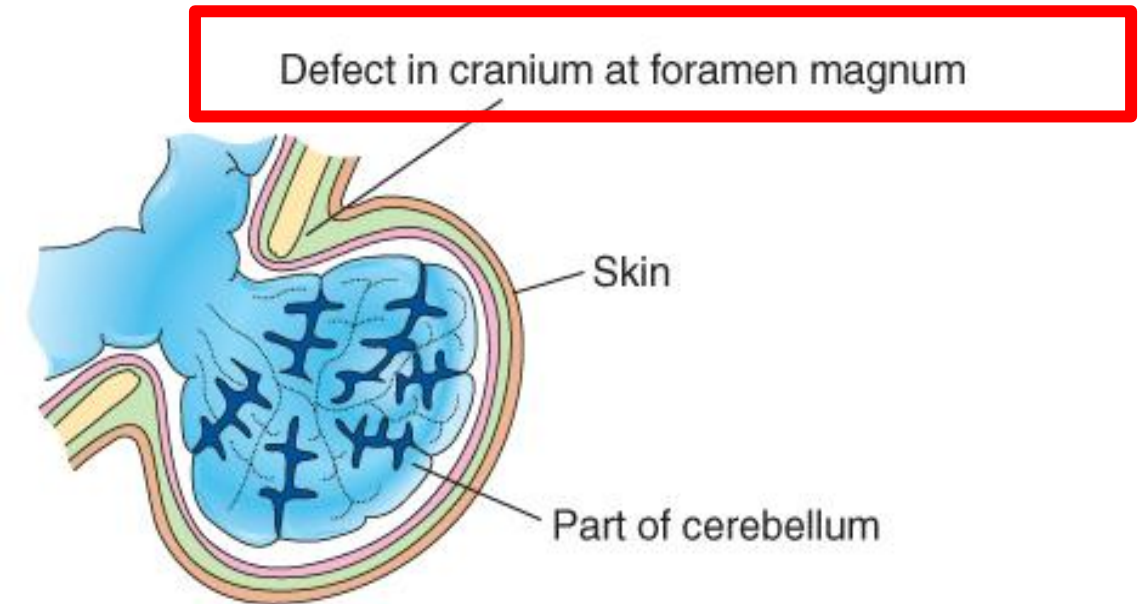
Swelling contains **meninges & CSF**



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Cranium Bifida with Meningoencephalocoele

Swelling contains **meninges, CSF & brain tissue**

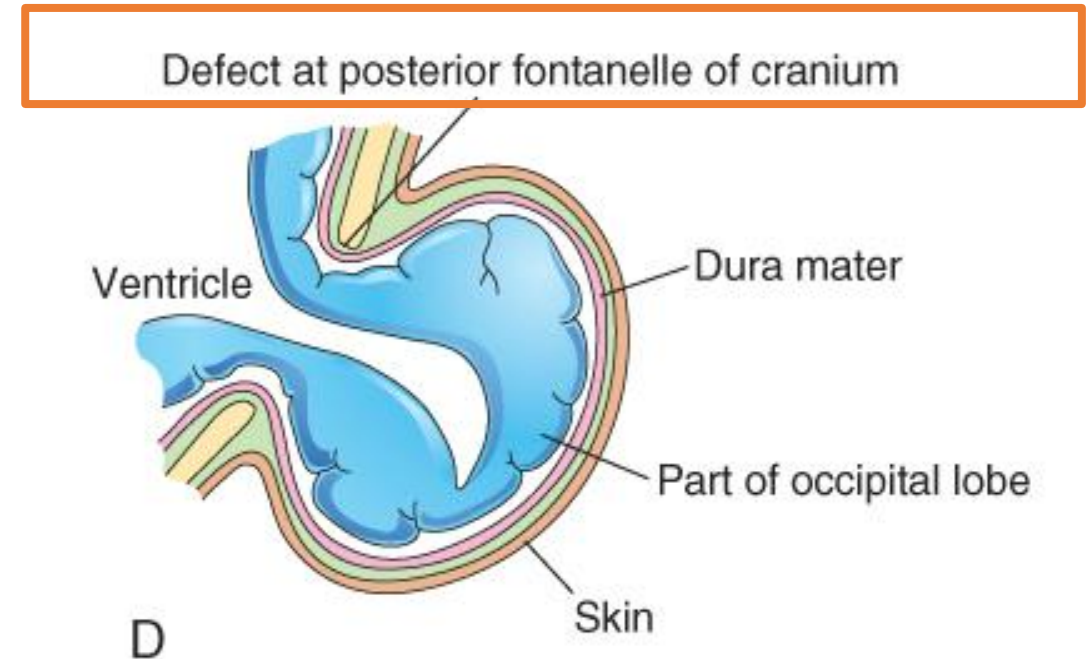


C

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Cranium Bifidum with Meningoencephalocele

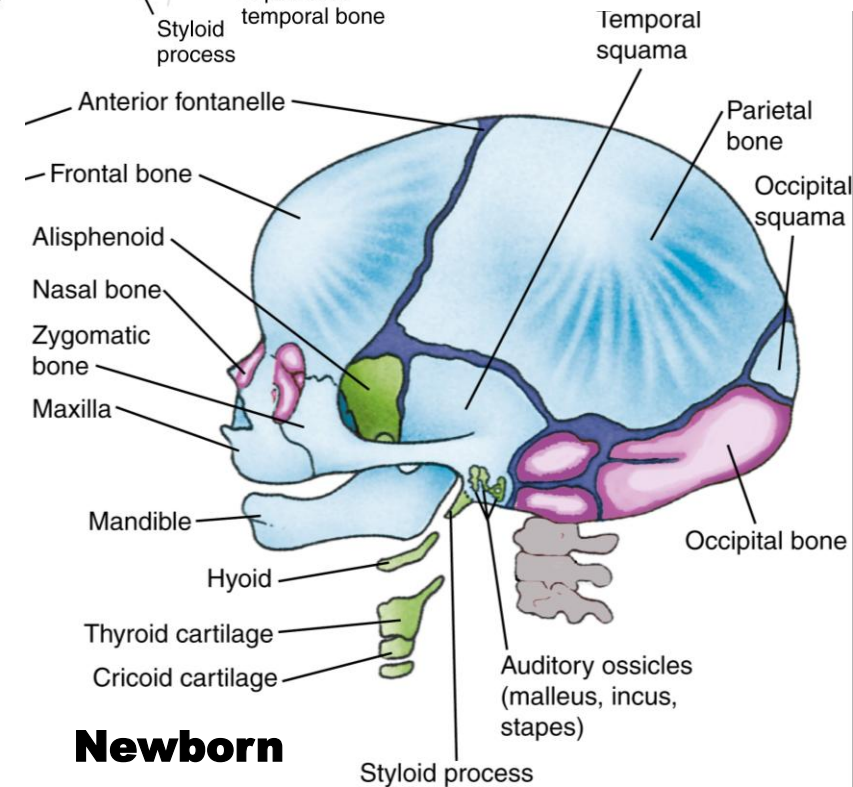
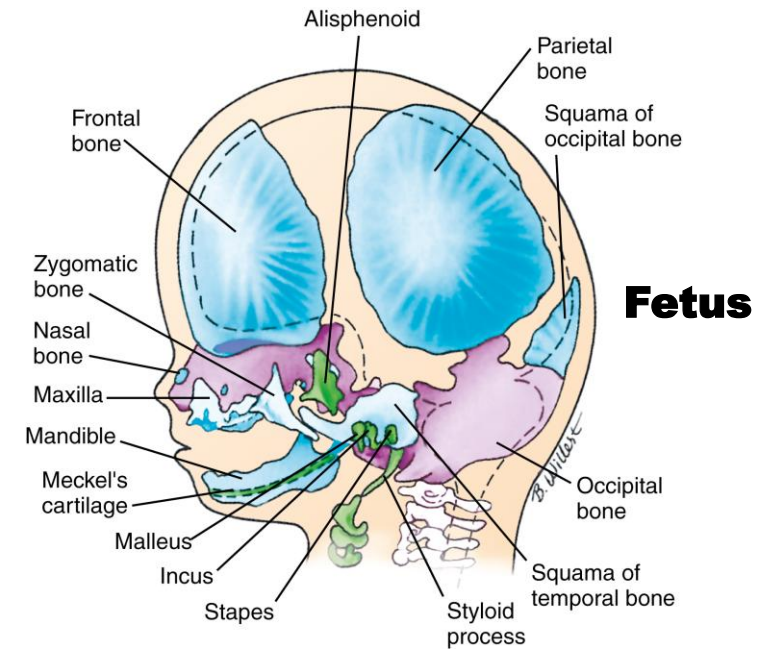
Swelling contains extensions of **meninges**, **CSF**, **brain tissue** and **ventricle**



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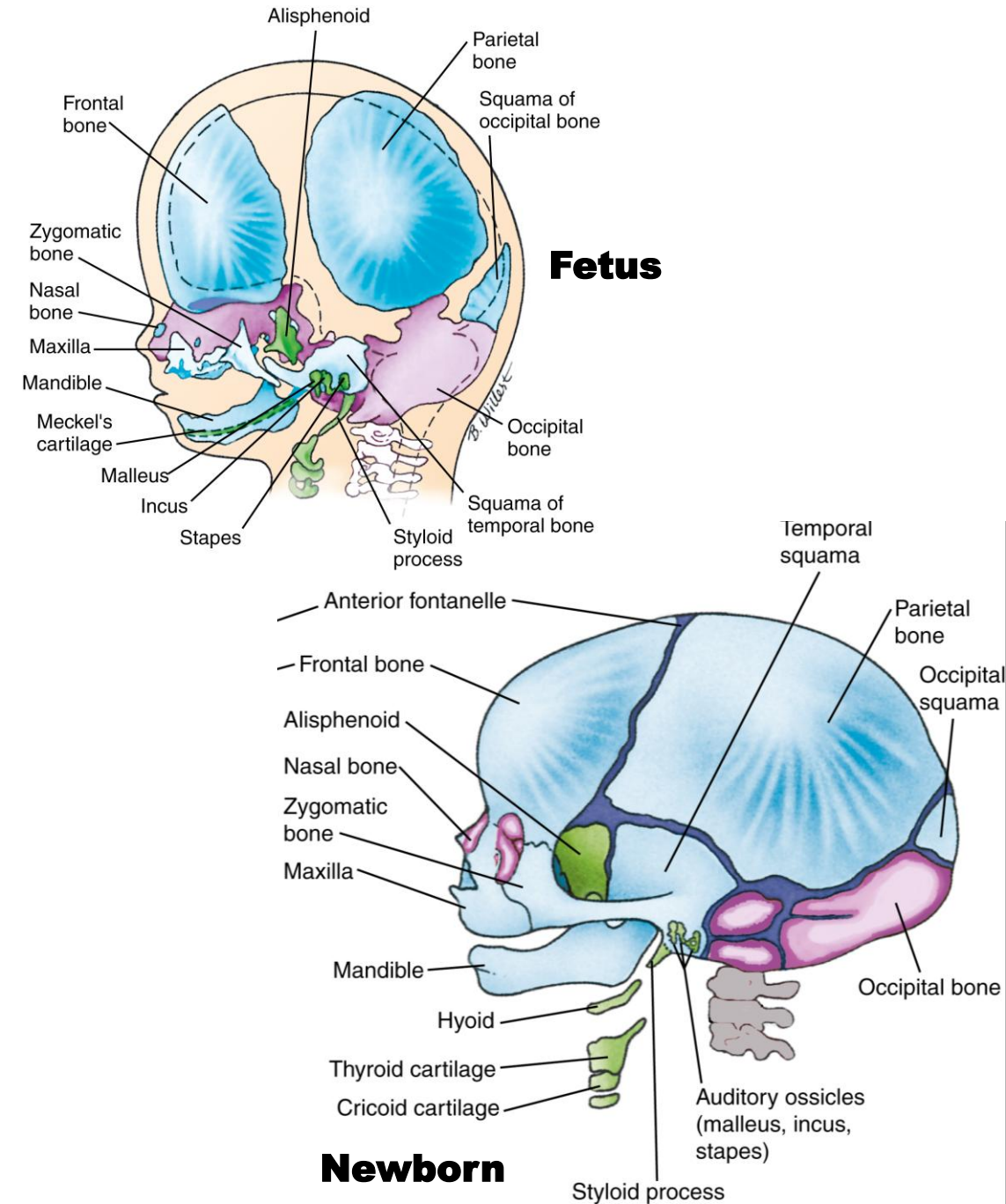
Membranous Viscerocranium

- **Intramembranous ossification** occurs in the maxillary prominence of the first pharyngeal arch
- It subsequently forms the **squamous temporal, maxillary** and **zygomatic** bones
- The squamous temporal bone becomes part of the neurocranium
- Mesenchyme in the mandibular prominence of the first arch condenses around its cartilage (Meckel cartilage) and undergoes intramembranous ossification to form the **mandible**



Cartilaginous Viscerocranium

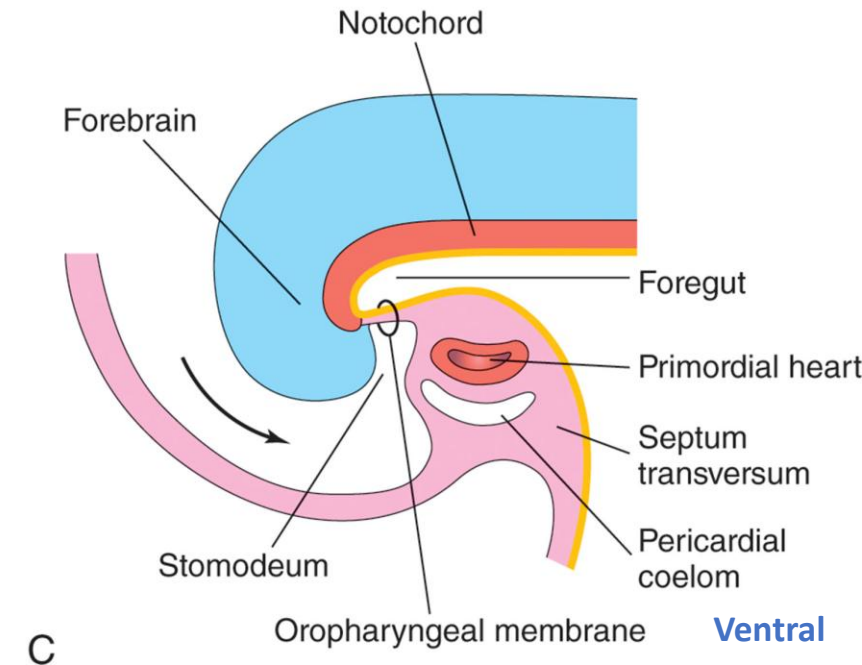
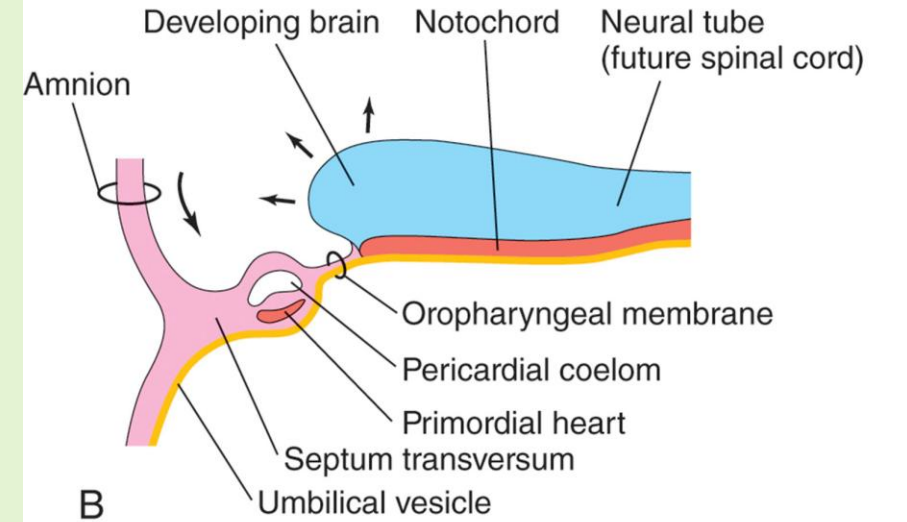
- The **middle ear bones, hyoid bone, styloid** process of temporal bone are derived from endochondral ossification of the pharyngeal arches cartilage



Clinical correlates	Embryological defects
Craniosynostosis	Premature closure of sutures
Trigonocephaly	Premature closure of frontal suture
Brachycephaly	Premature closure of coronal suture
Scaphocephaly	Premature closure of sagittal suture
Plagiocephaly	Asymmetrical premature closure of suture
Cranium bifidum	Midline defect in the formation of cranial vault

Folding of the cranial portion - headfold

- The developing brain expands rapidly and overgrows the underlying structures
- This creates rigidity in the dorsal region of the embryo
- Results in the more pliable (and less expanding) ventral region to be overgrown
- Pulls the cranial amnion with it

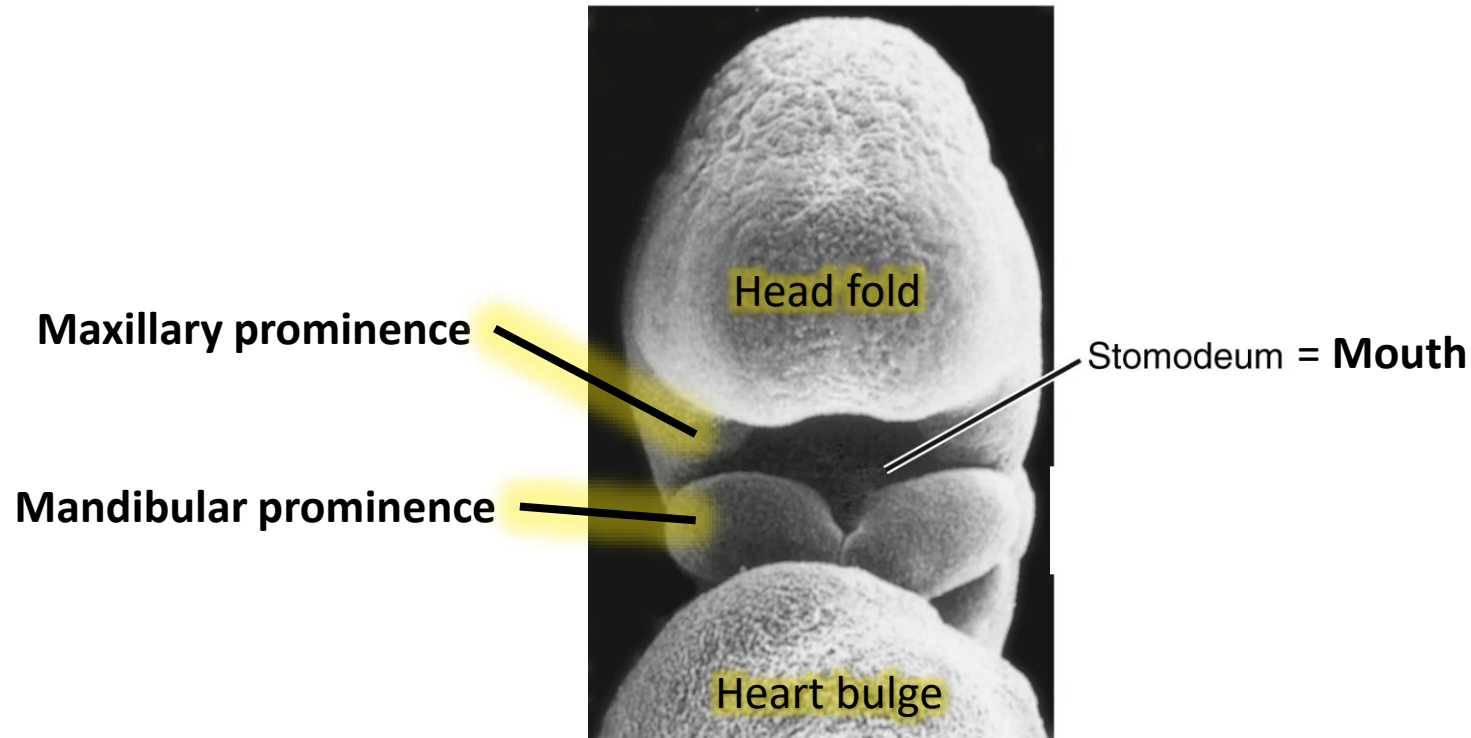


Dorsal

Ventral

Development of the Face: Formation of the Stomodeum

- Cranial - Frontal eminence
- Caudal- Heart prominence
- Lateral – Maxillary prominence
- Mandibular prominences fuse in the midline to form the caudal limit of the stomodeum
- Stomodeum is separated from the developing pharynx by the **oropharyngeal membrane**

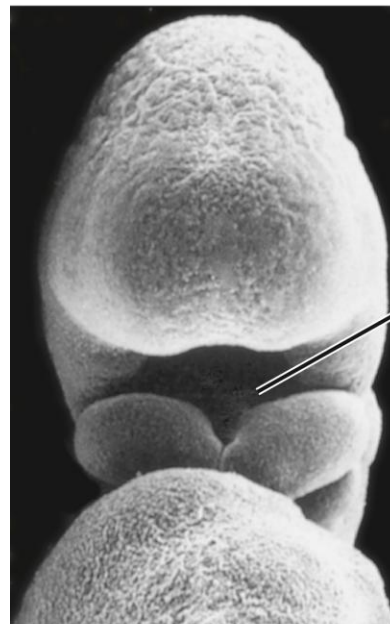


Development of the Face

Under influence of forebrain, frontonasal ectodermal zone and developing eye

Five facial primordia develop:

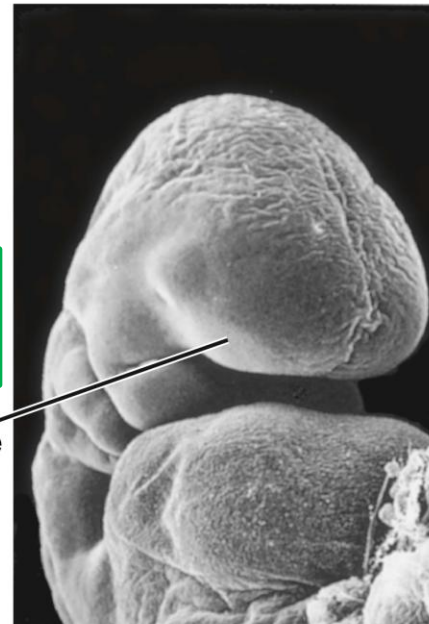
- One **frontonasal prominence** – surrounds the ventrolateral forebrain from which the optic vesicles form
 - **Paired maxillary** prominences
 - **Paired mandibular** prominences
- } Derived from 1st arch



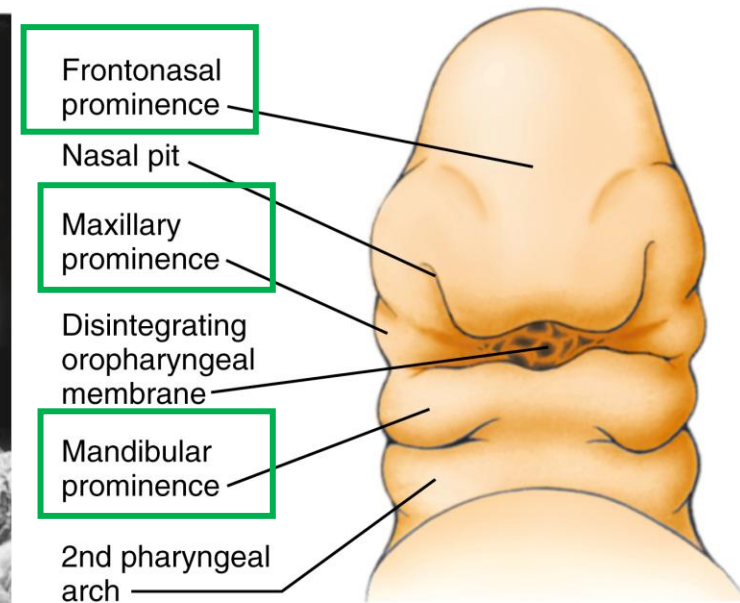
A 4th week

Stomodeum
Mouth

Nasal placode



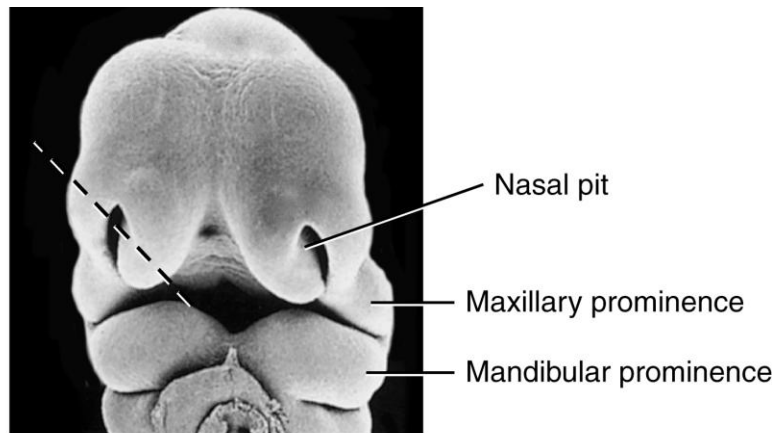
B 5th week



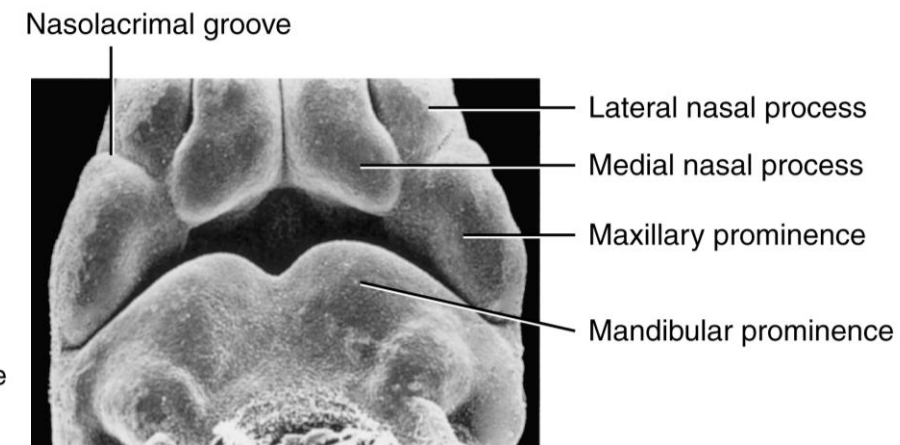
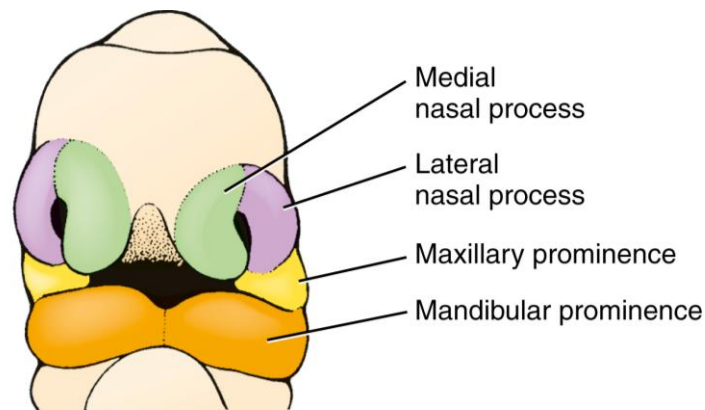
C 5th week

Development of Face

- Bilateral oval thickenings of surface ectoderm develop on the inferolateral parts of the frontonasal prominence - **Nasal placodes**
 - Mesenchyme in the margins of the placodes proliferate to produce **horseshoe shaped elevations**
 - Medial nasal prominence
 - Lateral nasal prominence
- Nasolacrimal groove is infero-lateral to the lateral prominence separating it from the maxillary prominence
- the different prominences of the face increase in size and change relative position



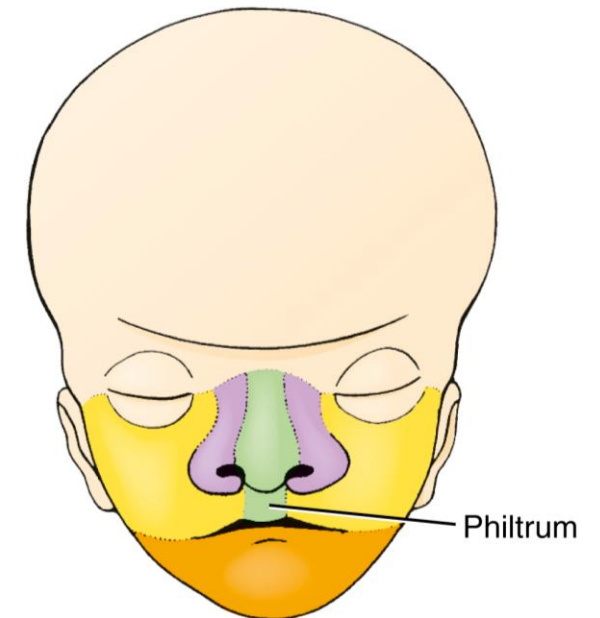
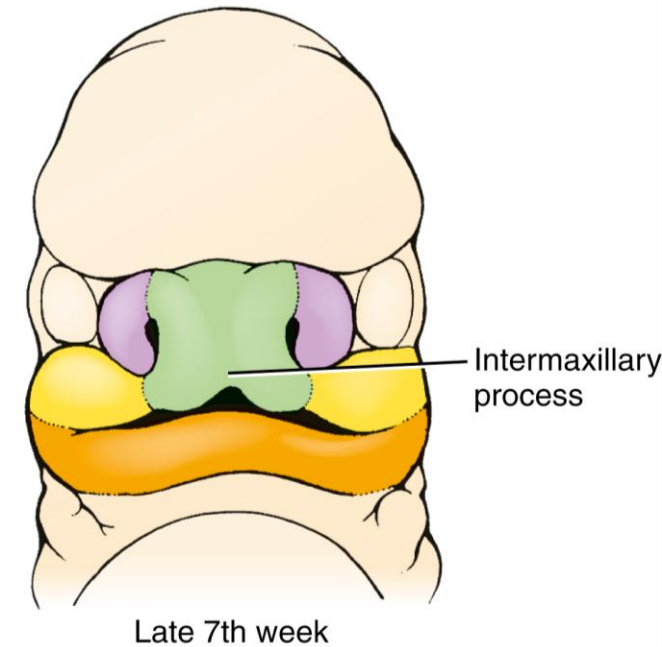
A Early 6th week



C Early 7th week

Development of Face cont.

- **Maxillary prominences** grow medially compressing the **medial nasal prominences** to the midline causing them to fuse
- Maxillary prominences increases in size to form the cheeks and maxilla
- The nasal pit deepens
- The nasolacrimal groove ectoderm forms a solid epithelial cord that detaches from the overlying surface ectoderm and runs from the medial corner of the eye to the lower aspect of the nasal cavity
- The **mandibular prominences merge** across the midline to form the lower lip and jaw



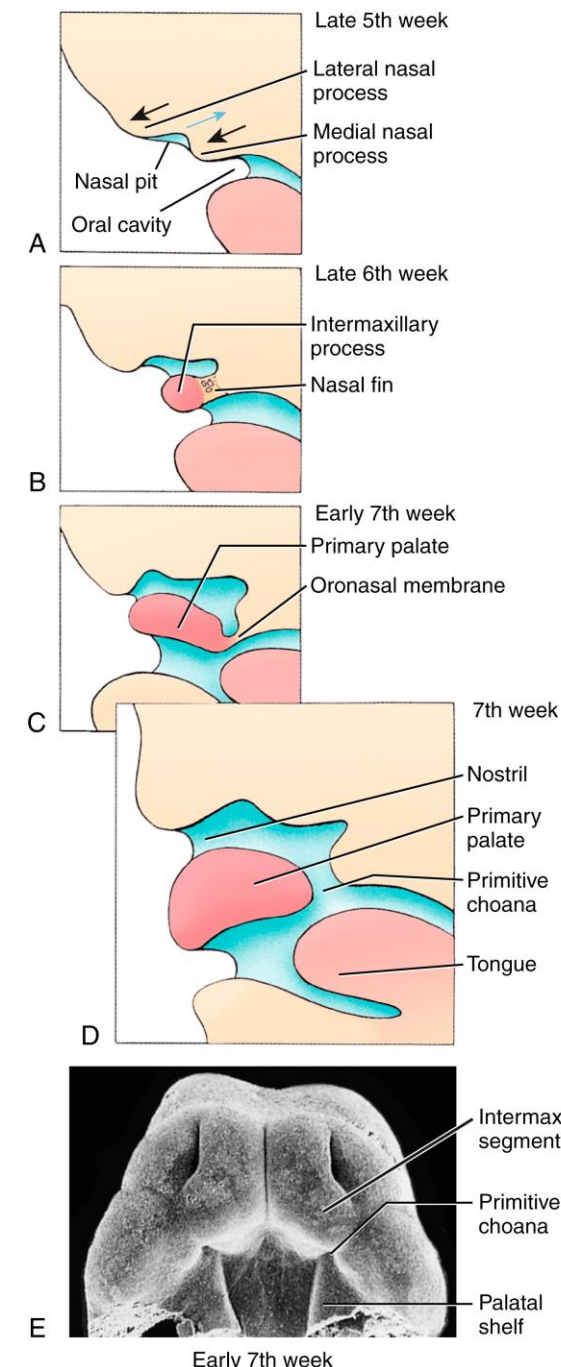
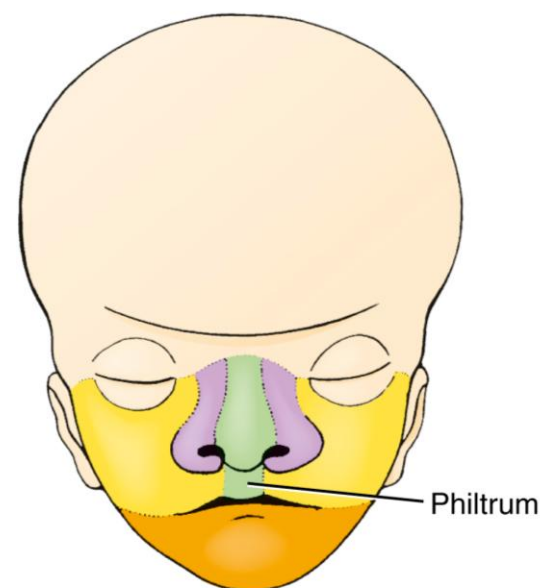
Formation of the intermaxillary segment

Merging of the two medial nasal prominences at a deeper level = **intermaxillary segment**.

Consists of:

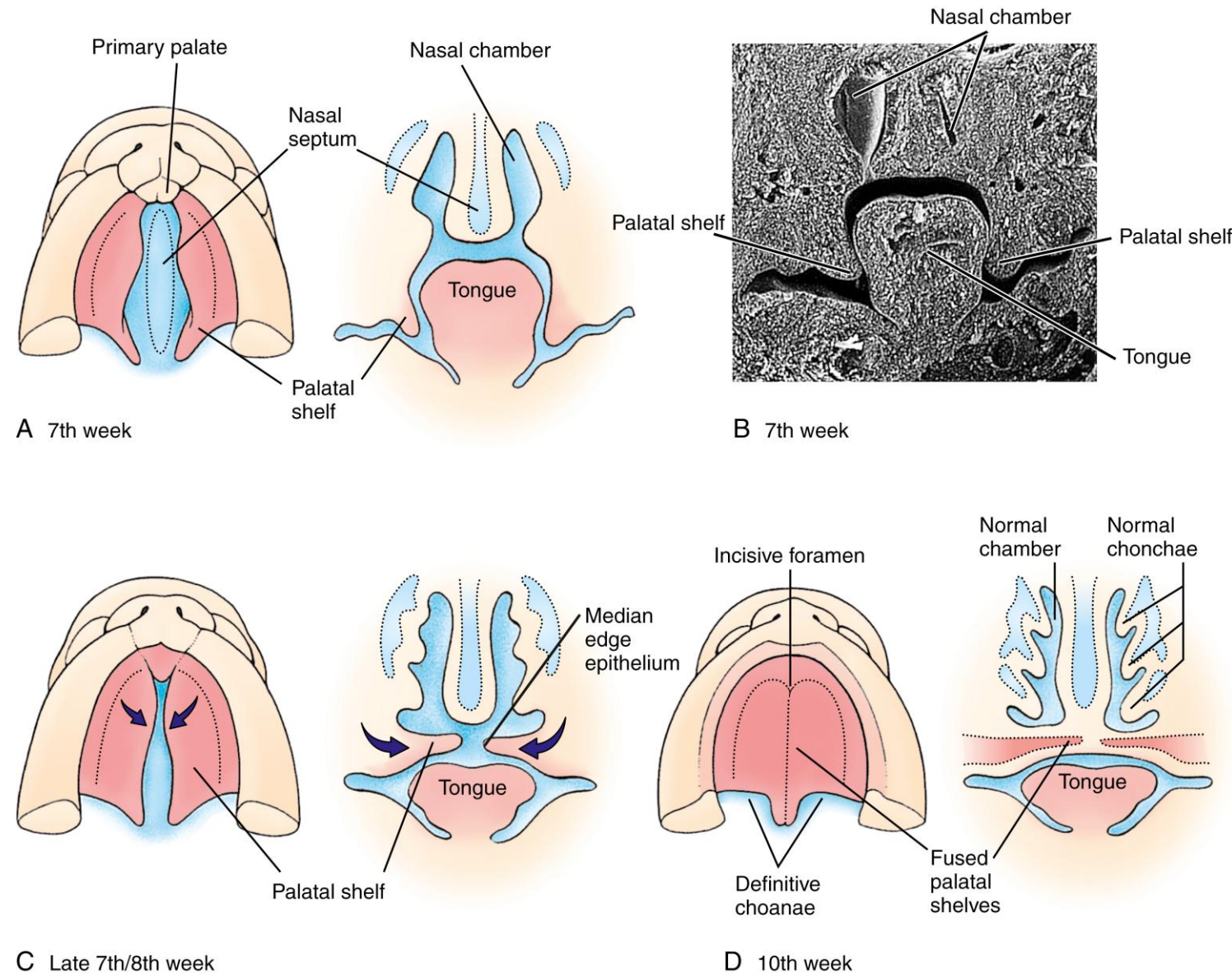
- Labial component forming the **philtrum** of the upper lip
- Upper jaw component which carries the **four incisor teeth**
- Palatal component which forms the **triangular primary palate**

The intermaxillary segment is continuous with the rostral portion of the nasal septum, which is derived from the frontal prominence.



Development of Secondary Palate

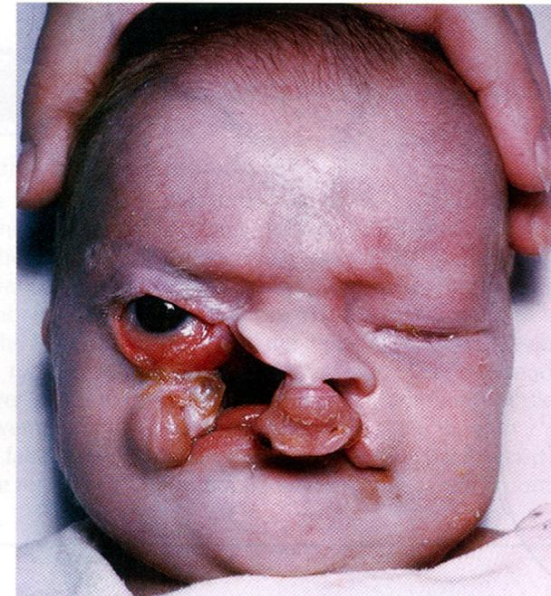
- Two **lateral vertical shelf like outgrowths** from the maxillary processes on either side of the tongue (lateral palatine processes)
- Reposition due to simultaneous growth of the lower jaw and processes = **horizontally placed**
- Nasal septum grows down simultaneous to fusion of palatine processes
- Fuse to the primary palate anteriorly and the junction is demarcated by the incisive foramen





Clefts

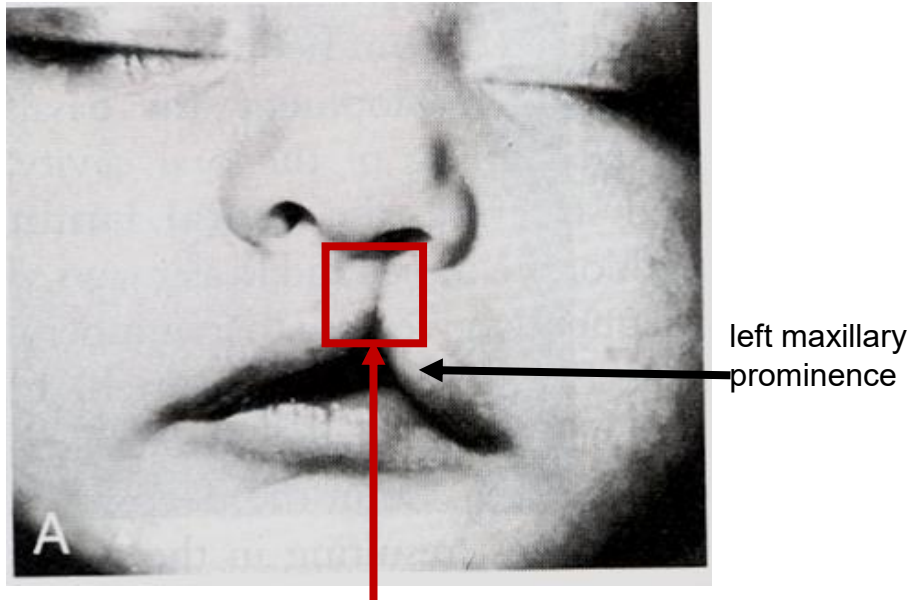
- Can occur on its own or as part of a syndrome
 - Multifactorial
 - Varies in severity
 - Lip and palate clefts are more common
-
- ❑ **Classified based on position to incisive foramen**
 - ✓ **Anterior**
Cleft lip, cleft of the primary part of the hard palate
 - ✓ **Posterior**
Cleft of the secondary part of hard palate, soft palate
 - **Facial clefts**



Cleft Lip

- Cleft lip can occur independently from underlying bony deformities
- Can be unilateral or bilateral

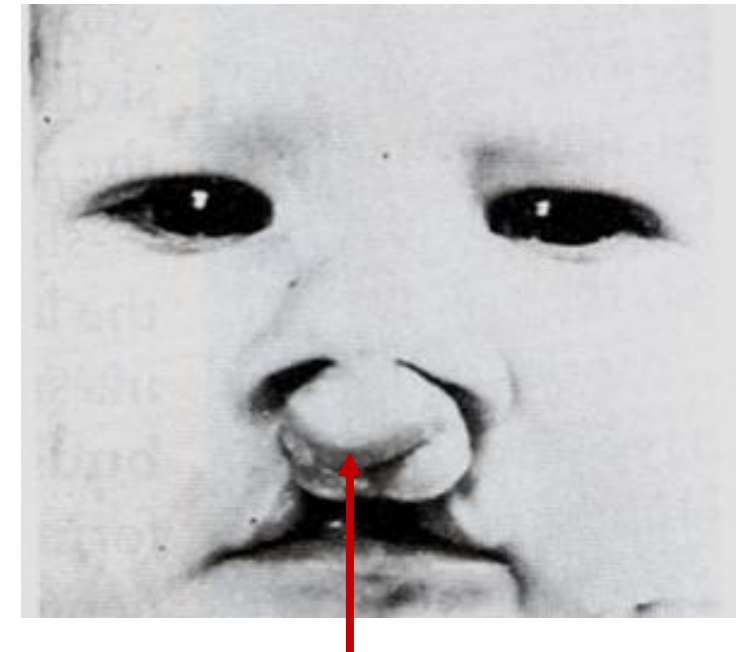
Unilateral Cleft Upper Lip



Intermaxillary segment

Failure of fusion of the **left maxillary prominence** with the intermaxillary segment

Bilateral Cleft Upper Lip



Intermaxillary segment

Failure of **both maxillary prominences** to fuse with the intermaxillary segment



Median Clefts

- Rare
- Upper may be associated with varying degree of loss of midline structures e.g., holoprosencephaly, single incisor teeth, nasal septum absent, single eye

Upper Lip



Failure of fusion of the two medial nasal prominences to form the intermaxillary segment

Lower Lip



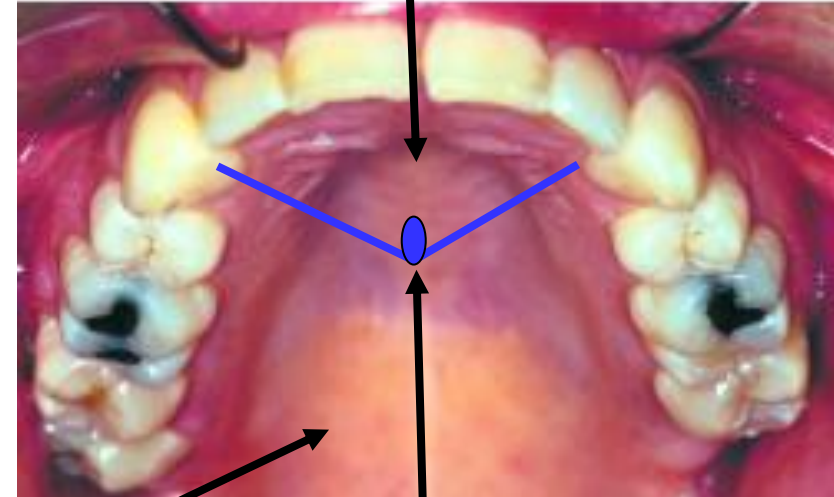
Failure of fusion of the mesenchyme of the mandibular prominences

Cleft Palate

- Classification with reference to incisive fossa
 - Anterior or posterior
- 1: 2500 births; more common [65%] in females
- Not related to maternal age

Adult structure

Primary palate (Intermaxillary segment)

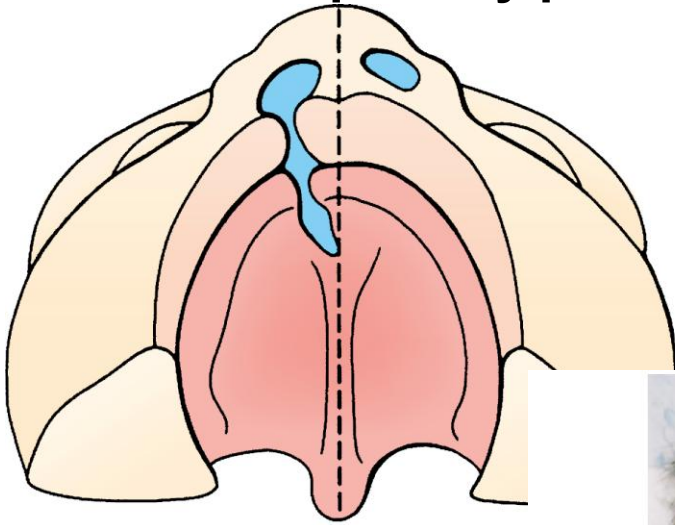


Secondary palate
(Lateral palatine processes)

Location of incisive fossa

Clefts of the Anterior Palate

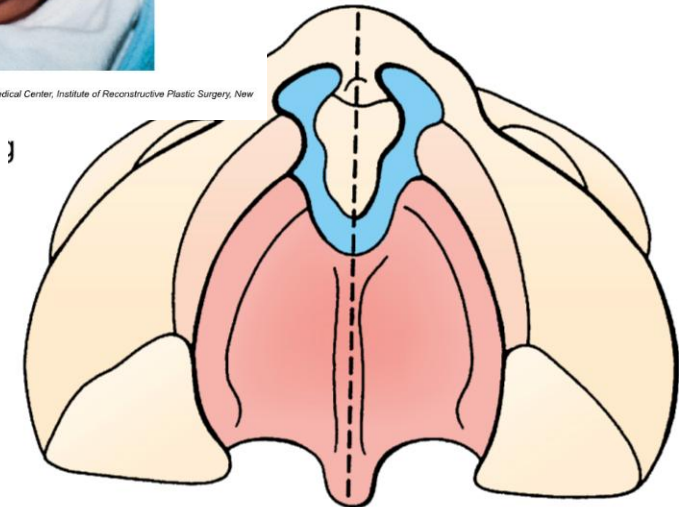
- Anterior to the incisive fossa
- May involve the lip or not
- May be bilateral or unilateral
- Results from failure of the **fused lateral palatine processes (secondary palate)** to meet and **fuse with the primary palate (intermaxillary segment)**



Courtesy Dr. A. E. Chudley, Section of Genetics and Metabolism, Department of Pediatrics and Child Health, Children's Hospital, Winnipeg, Manitoba, Canada.

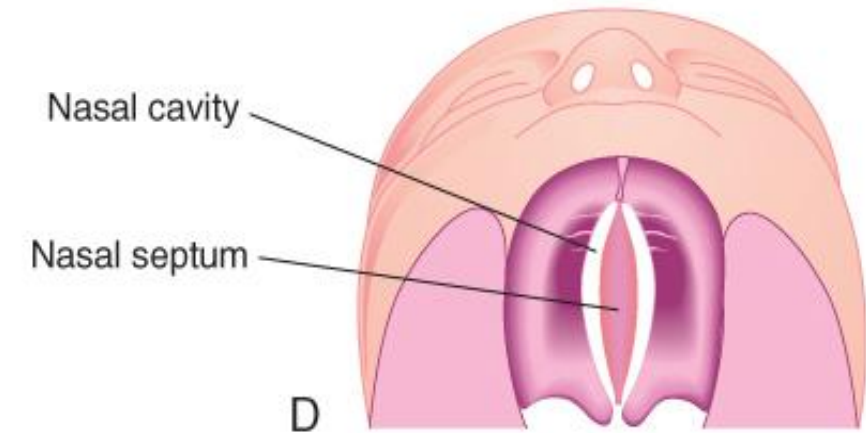
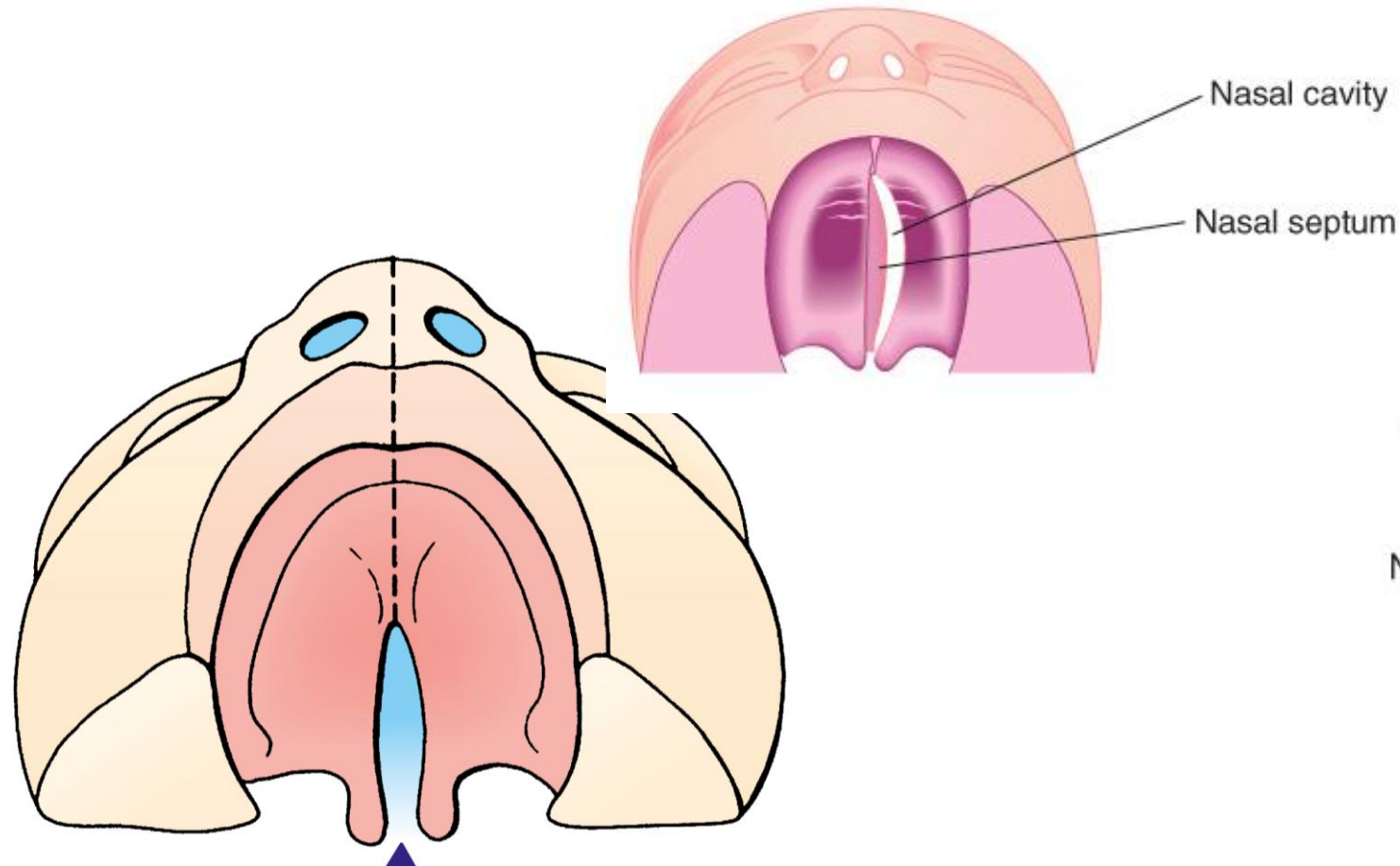


Courtesy Dr. Barry H. Grayson and Dr. Bruno L. Vendittelli, New York University Medical Center, Institute of Reconstructive Plastic Surgery, New York, NY.

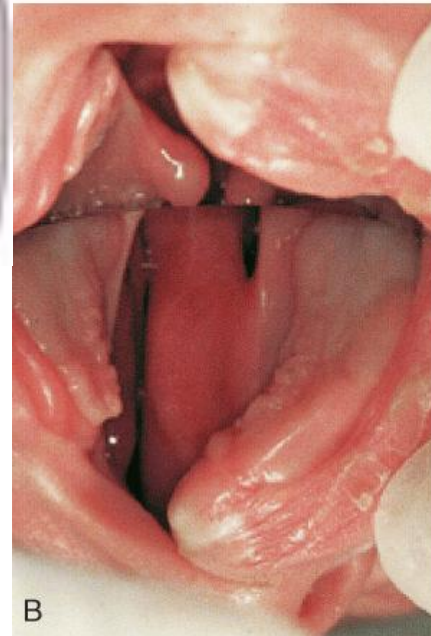
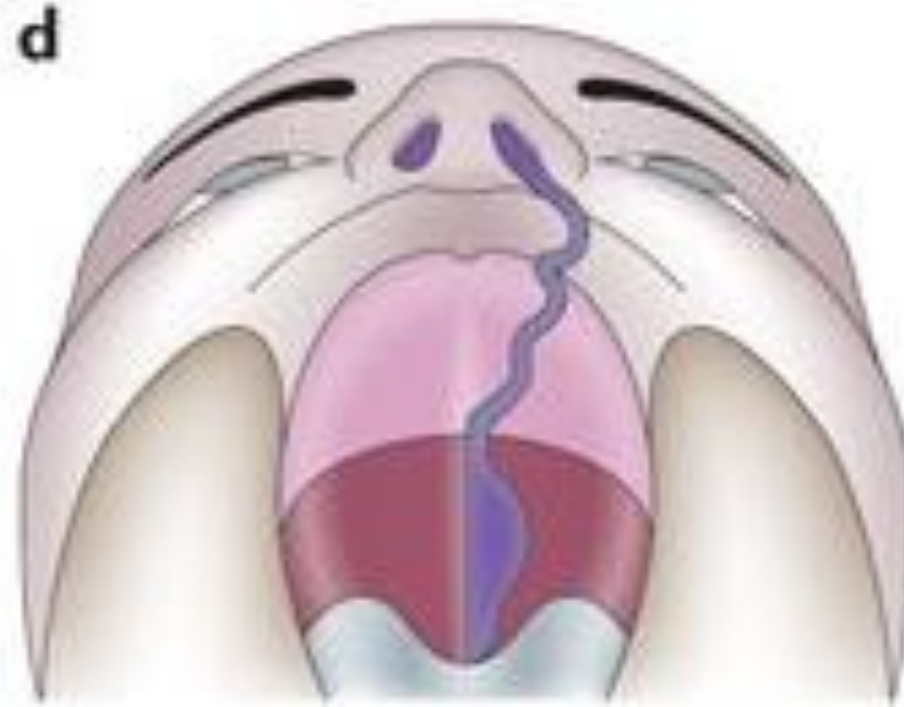


Clefts of the Posterior Palate

- Posterior to the incisive fossa
- Can be isolated from an anterior or not
- Results from failure of the **lateral palatine processes to meet and fuse with each other and the nasal septum**



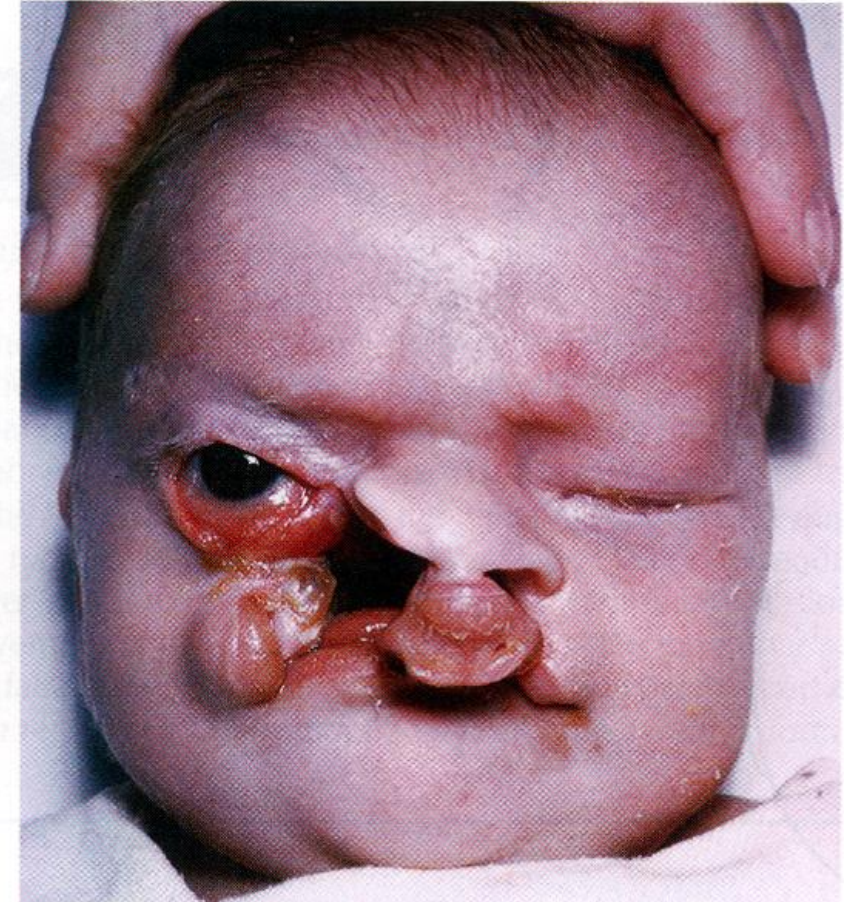
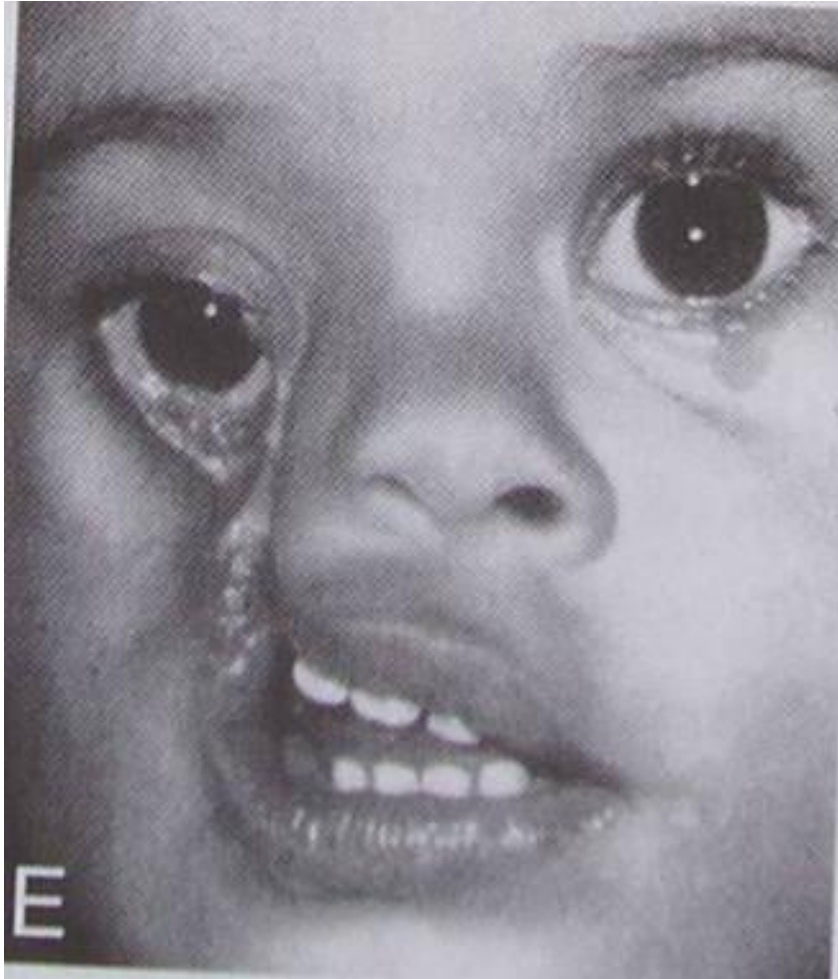
Complete cleft lip and palate



Courtesy Dr. John B. Mulliken, Children's Hospital Boston, Harvard Medical School, Boston, MA.

Oblique Facial Cleft

Failure of **maxillary prominence** to fuse with **lateral nasal prominence**
Nasolacrimal duct is visible.



Right sided oblique facial cleft
and left cleft upper lip

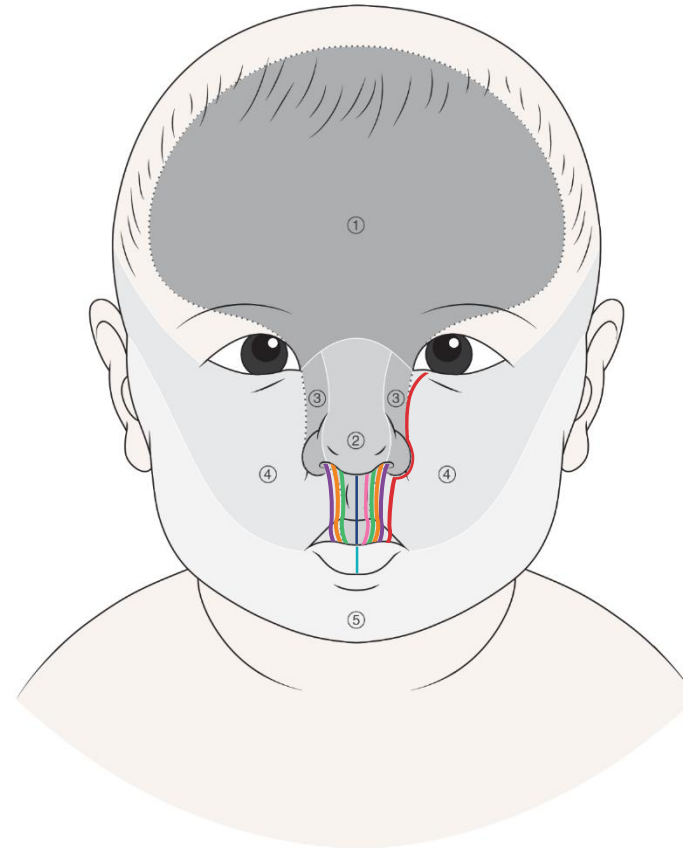
CLEFTS:

Facial Prominences:

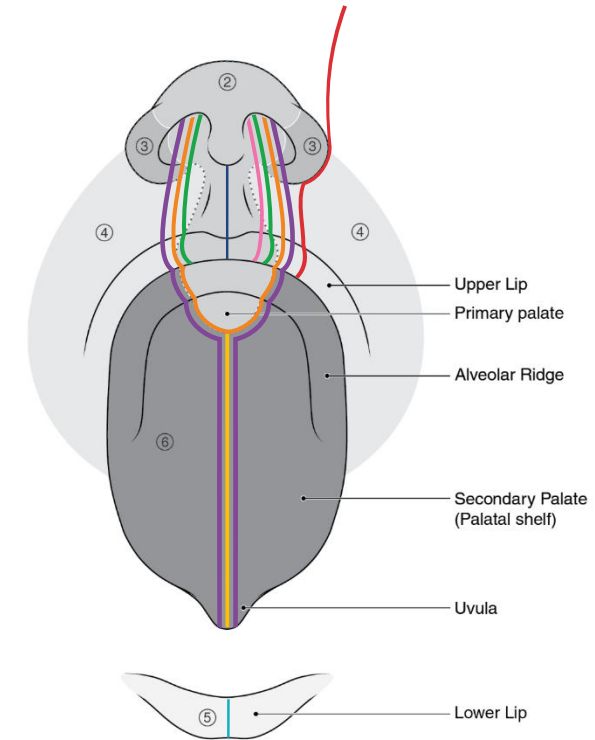
- Frontonasal Process
- ① Frontal Prominence
- ② Fused Medial Nasal Prominence (Intermaxillary Segment)
- ③ Lateral Nasal Prominence
- ④ Maxillary Prominence
- ⑤ Mandibular Prominence
- ⑥ Palatine Shelf

Type of Cleft:

- Unilateral Cleft Upper Lip
- Bilateral Cleft Upper Lip
- Median Cleft Upper Lip
- Median Cleft Lower Lip
- Anterior Cleft Palate
- Posterior Cleft Palate
- Anterior and Posterior Cleft Palate
- Oblique Facial Cleft



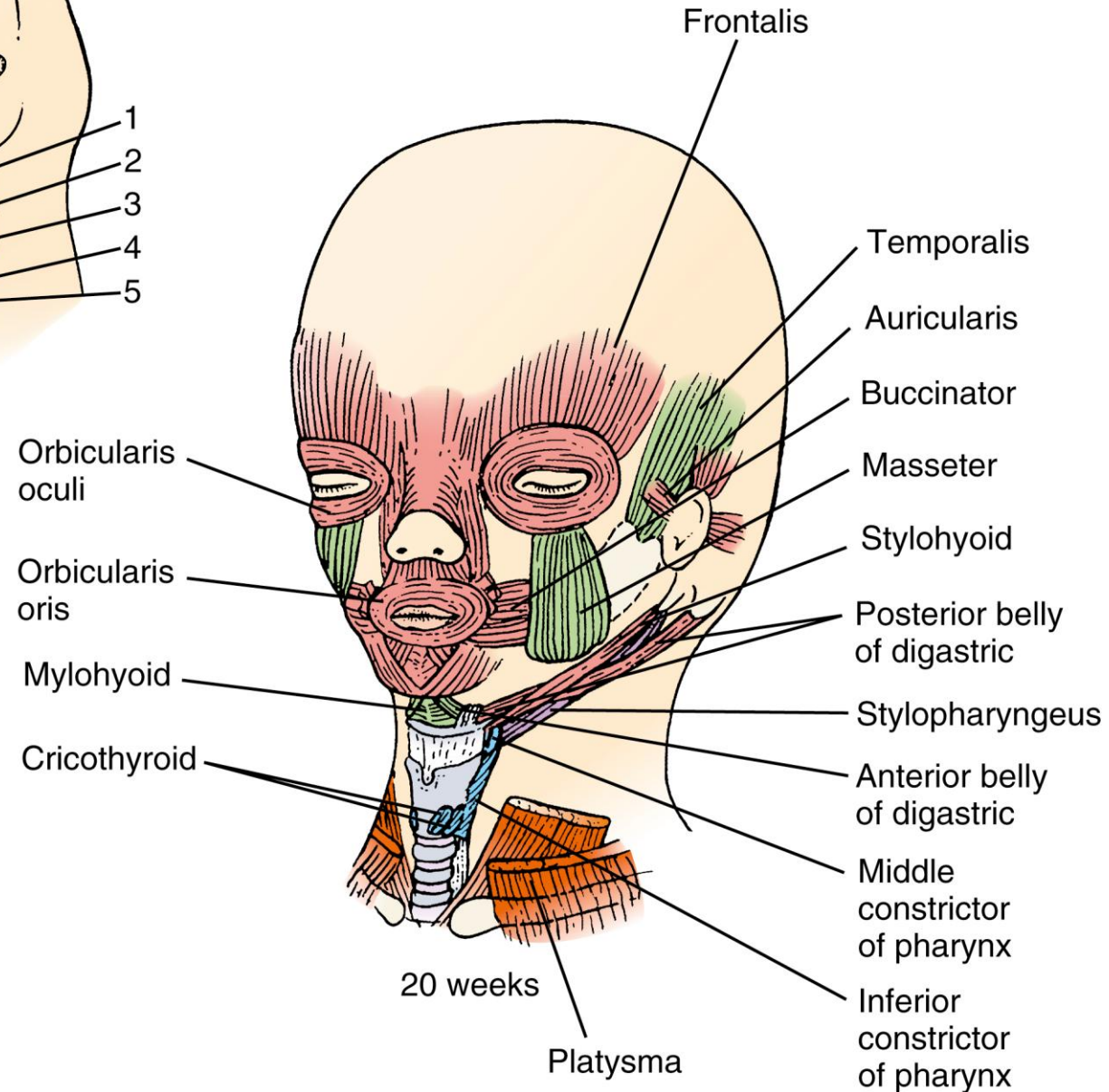
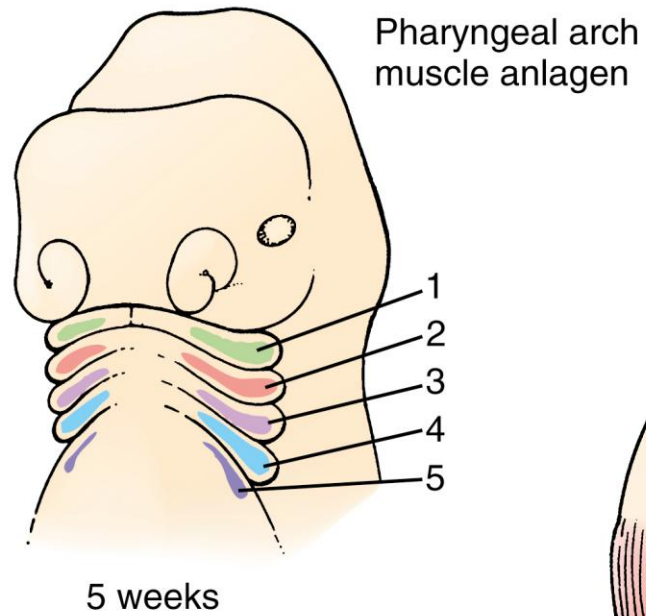
Anterior view of Face



Inferior view of Palate

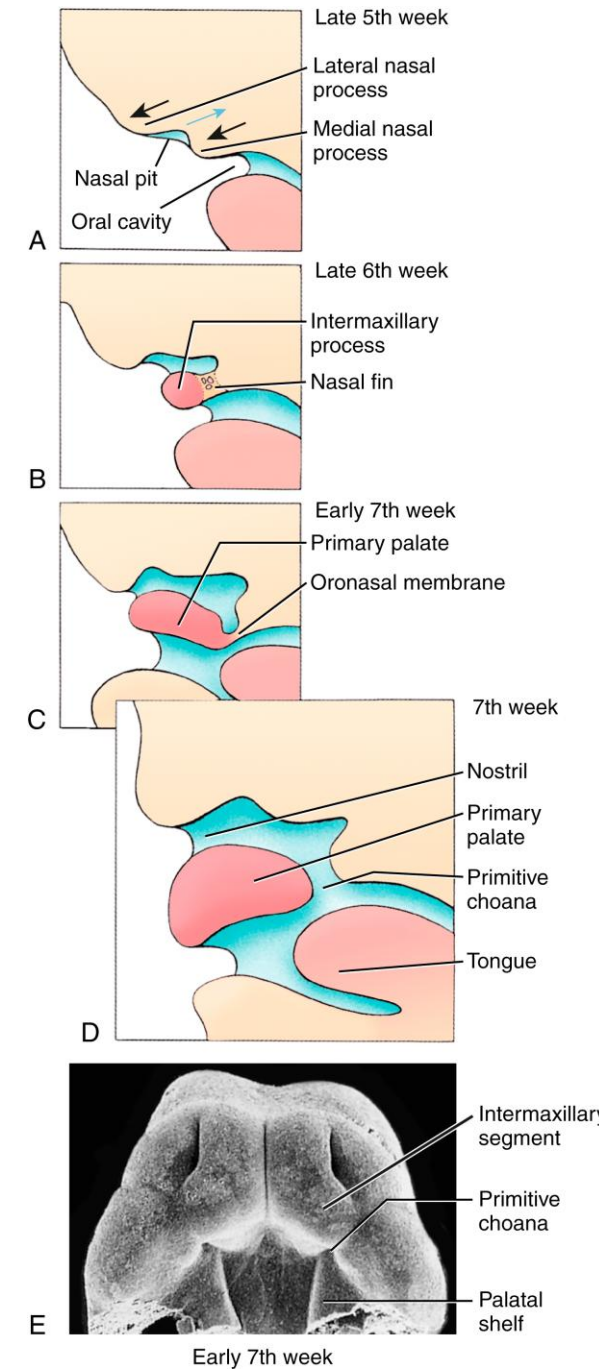
Clinical correlates	Embryological defects
Cleft upper lip	
Unilateral cleft upper lip	Failure of fusion of the left/right maxillary prominence with the intermaxillary segment
Bilateral cleft upper lip	Failure of fusion of both maxillary prominence with the intermaxillary segment
Median cleft upper lip	Failure of fusion of the two medial nasal prominences to form the intermaxillary segment
Median cleft lower lip	Failure of fusion of the mesenchyme of the mandibular prominences
Cleft palate	
Cleft anterior palate	Failure of the secondary palate to meet and fuse with the primary palate
Cleft posterior palate	Failure of the lateral palatine processes to meet and fuse with each other and the nasal septum
Cleft anterior and posterior palate	Failure of the lateral palatine processes to meet and fuse with the primary palate, with each other, and the nasal septum
Oblique cleft lip	Failure of maxillary prominence to fuse with lateral nasal prominence

Muscles



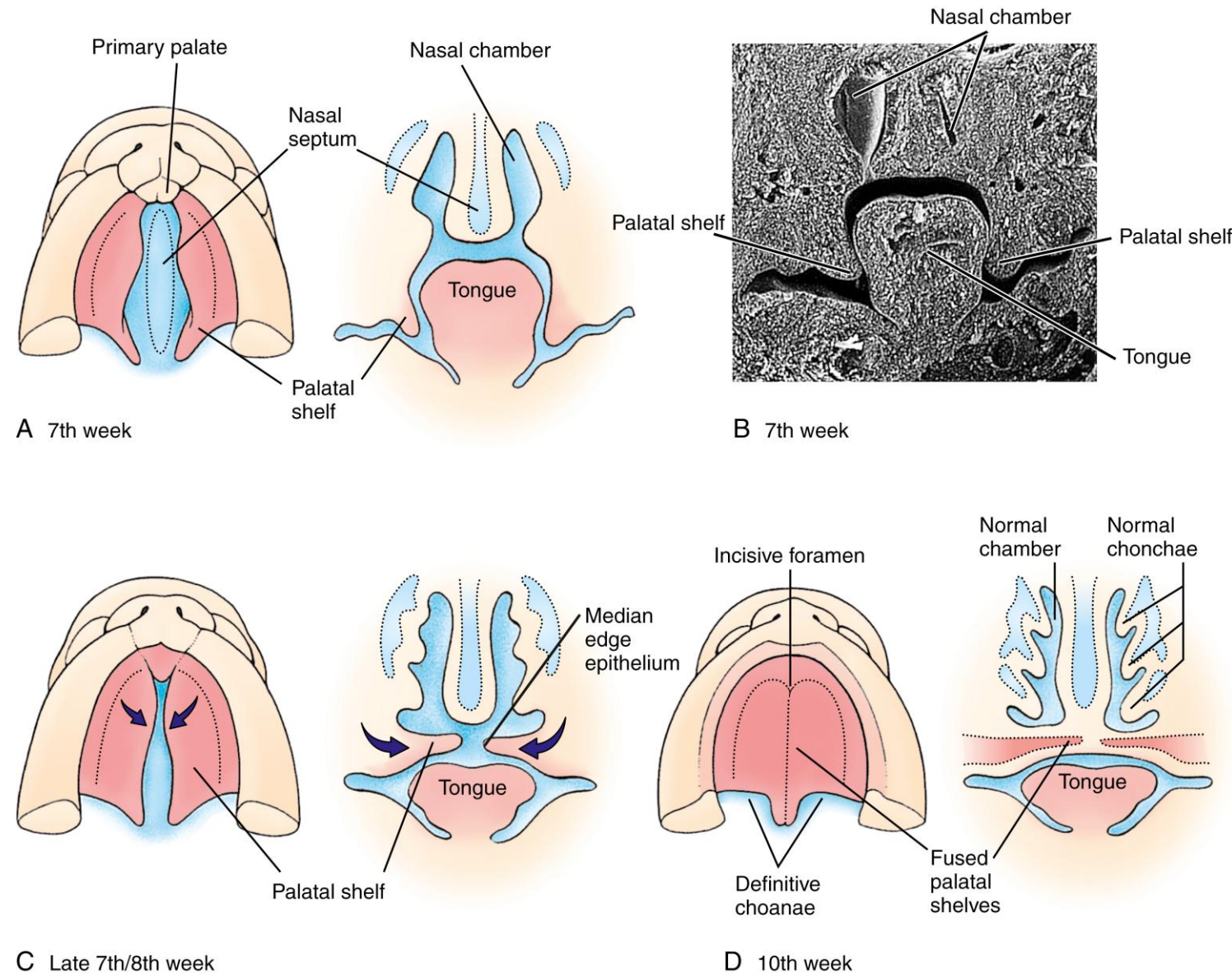
Nasal Cavities

- Nasal pit deepens due to **overgrowth of nasal prominences** and penetration into underlying mesenchyme
- Separated from the developing oral cavity by the oronasal membrane
- Oronasal membrane starts to rupture forming the future choana
 - Choana at first lie on either side of the midline
 - move backward as the nasal cavity increases in size and the definitive palate is formed

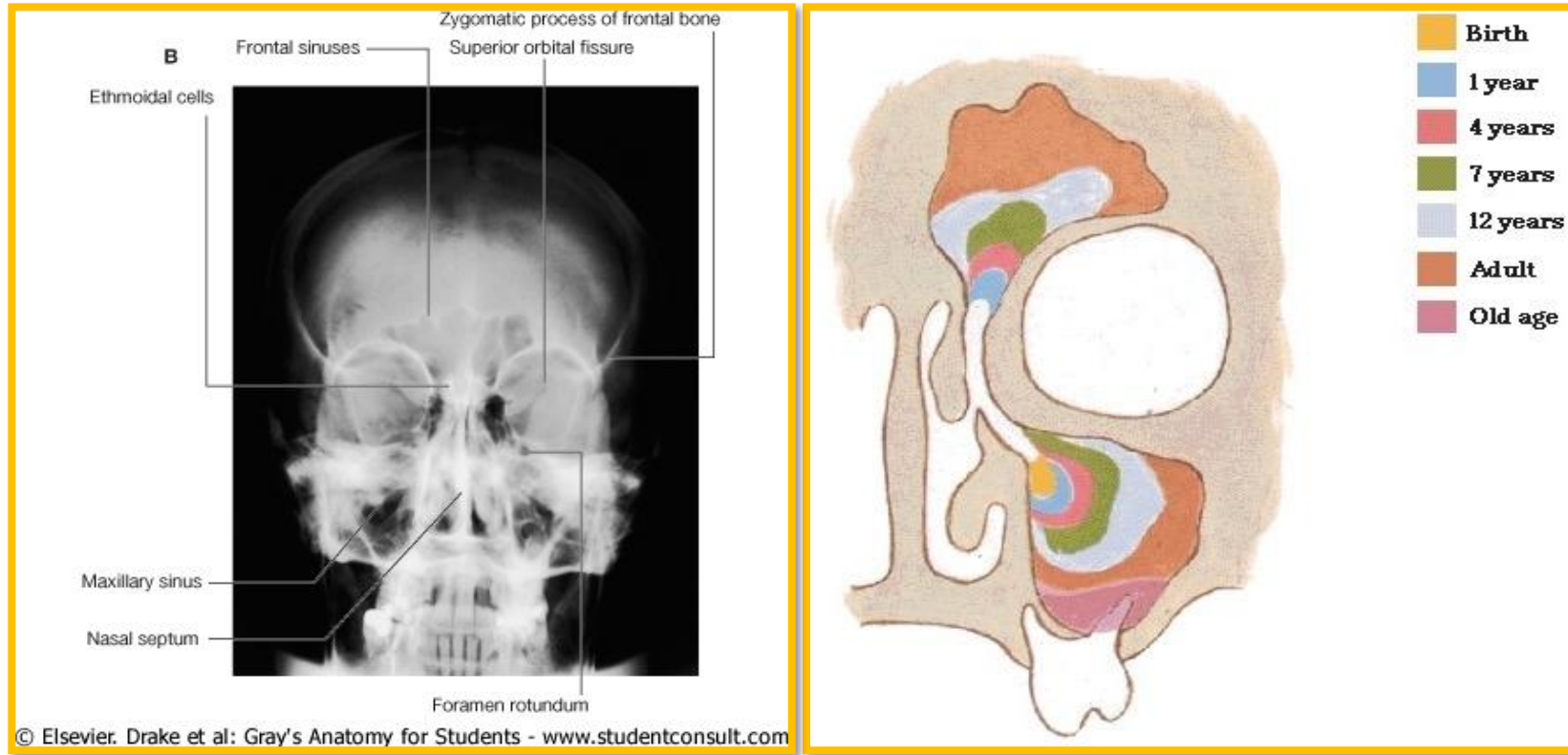


Nasal Cavities

- The fused maxillary prominences give rise to the **nasal septum**
- The **nasolacrimal duct undergoes canalization**, and the proximal end distends to form the lacrimal sac.
- Epithelial plug forms from growth of surrounding tissue and separates the nasal cavity from the amniotic cavity



Paranasal sinuses



They develop as diverticula of the lateral nasal wall and extend into the maxilla, ethmoid, frontal and sphenoid bones.

They reach their maximum size during puberty and contribute to the definitive development of the face.

END